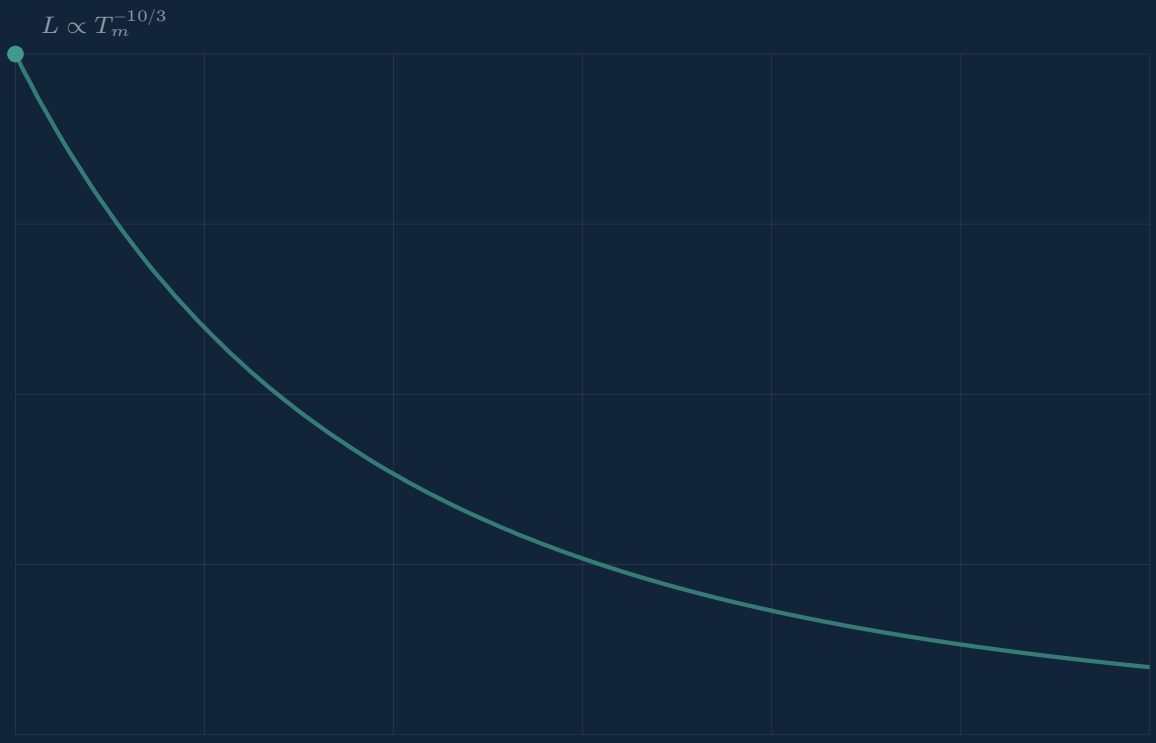
TECHNICAL WHITE PAPER · PRINCIPLES AND METHODOLOGY · ENGLISH EDITION

DTM

Reducer Health Monitoring Principles and Methodology

The decision methods explained from mathematics and physics

Starting from the torque balance of a robot joint and the power law of reducer friction and life, this paper derives the formulas, statistical methods and assumptions behind every decision step in DTM: friction decomposition, baselines and false-alarm control, multi-evidence fusion, Weibull life, damage attribution to program lines, load reduction by time scaling, and verification of improvements.



Domingo He October 2026 For DTM v2.1.0 (Beta)

I design, AI writes the code, I verify | Apache-2.0 open source

All numerical examples in this paper are illustrative or from demonstration data

How to read this paper

This paper explains the mathematics and physics behind every decision step of DTM (Disturbance Torque Monitor). DTM is open-source software that monitors the health of industrial robot reducers. It reads the "disturbance torque" that the robot controller already computes, so no sensors need to be added and production does not have to stop. It is used to detect reducer wear early and, in the advanced module (Pro), to find which line of the robot program is consuming reducer life.

How this paper is written

- Each section first explains the **physical or statistical reasoning**, then gives the **formulas and derivations**, and finally describes **what DTM actually does** (blue box "DTM implementation"). The two are written separately so that it is easy to tell which parts are principles and which are engineering choices.
- The thresholds and coefficients in the text (for example 1.5 times, 0.025/°C) are all **starting values**, based on the author's maintenance experience and public data. They can be adjusted in the configuration file and have not yet been validated on a production line over the long term (see Section 9).
- All numerical examples are illustrative values chosen to explain the formulas, or come from the demonstration data built into the software; none of them are measurements from any factory.

Readers and prerequisites

The paper is written for maintenance engineers, equipment engineers and researchers who want to check the methods. Required background: calculus, basic probability and statistics, and the concepts of rigid-body dynamics. The chapters are largely independent and can be read selectively:

If you want to know	Read
Why reducers fail even though every move is "within the allowable range"	1 Physics of reducer failure and life
How disturbance torque becomes "friction"	2 Measurement principle
How to detect "anomalies" without false alarms	3 Baselines, statistics and false-alarm control
Why replacement requires two kinds of evidence	4 Multi-evidence fusion
How life and spare parts are calculated	5 Weibull life statistics
Which program line is wearing the reducer	6 Program-level damage attribution
How to reduce load without changing cycle time	7 Time-scaling theory
How to prove an improvement works	8 Verification methods

Relation to other documents

This paper focuses on "why it is calculated this way". The complete parameters, variants and the legal statement of disclosure are in the *Technical Disclosure* in the source code (`docs/TECHNICAL_DISCLOSURE.md`, English edition `docs/TECHNICAL_DISCLOSURE.en.md`); the completion and verification level of each feature is in the *Feature Status* table (`docs/FEATURE_STATUS.md`); what is deliberately not done is in the *Design Constraints* (`docs/DESIGN_CONSTRAINTS.md`).

Relation to the manufacturer's functions

The robot manufacturer also offers reducer monitoring; for example, FANUC's ZDT catalog lists per-axis reducer deterioration diagnosis, motor torque analysis and grease-change notification [19]. DTM is not the first system for predictive maintenance of reducers and is not meant to replace the manufacturer's functions. Its focus is: (1) the decision methods are fully published and can be checked and reproduced; (2) load is attributed to program lines and reducer angles; (3) the effect of program changes is proven by measurements with a control group.

Main symbols

Symbol	Meaning	Unit
$q,\ \dot{q},\ \ddot{q}$	Joint angle, angular velocity, angular acceleration	°, °/s, °/s^2
ω	Angular velocity at the reducer output	rad/s
τ_{dis}	Disturbance torque (actual torque estimated by the controller minus model torque)	% of rated or N m
T	Torque transmitted by the reducer	N m
$T_0,\ N_0,\ L_0$	Rated torque, rated output speed and rated life of the reducer	N m, rpm, h
p	Life exponent ($10/3$ for RV type and roller bearings, 3 for ball bearings and gearboxes)	—
$\tau_+,\ \tau_-$	Disturbance torque at the same position during constant-speed forward and reverse runs	as τ_{dis}
$F,\ G$	Friction; direction-independent offset (gravity, balancer, payload setting)	as above
$F_c,\ F_v$	Constant (Coulomb) friction; viscous friction coefficient	as above; as above per speed
$r,\ z$	Ratio to baseline; z-score	—
k	Temperature coefficient	1/°C
s	Severity (0–1.5; 0.5 = caution, 1.0 = alert)	—
$c,\ W$	Iron-debris concentration; iron-debris generation rate	wt%, mg/h
$\beta,\ \eta$	Weibull shape parameter; scale parameter	—, h
D	Damage (equivalent revolutions at rated torque)	rev
σ	Time-scaling factor (motion time becomes σ times longer)	—
θ	Reducer output angle	°

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1 Why reducers fail: the physics of reducer failure and life

Selection calculations for robot joint reducers usually check only two things: whether the torque of each motion stays below the allowable value, and whether the rated life at the average load is sufficient. What is often seen in the field, however, is that reducers "stay within the allowable range every time and still fail again and again". Starting from the power law of fatigue life, this chapter explains why the number of cycles, load concentration and temperature matter more than a single peak. The decision methods of the following chapters are built on these relationships.

1.1 Structure and failure modes of RV reducers

An RV-type (cycloidal-pin) reducer has two stages. In the first stage the input gear drives several spur gears. In the second stage the crankshafts (eccentric shafts) driven by the spur gears move the cycloidal discs, which mesh with a ring of pins inside the housing, and the output is supported by a large main bearing. The load passes in turn through the gear tooth contacts, the needle bearings on the crankshafts, the contacts between the cycloidal discs and the pins, and the main bearing. All of these are **rolling or rolling-sliding contacts**, and the main failure mechanisms are contact fatigue and wear:

- **Lubrication deterioration**: grease oxidizes, oil separates, wear particles or water get in, and the oil film becomes thinner; contact shifts from fluid-film to boundary lubrication, friction rises and wear accelerates.
- **Surface fatigue**: repeated contact stress creates subsurface cracks which grow into pitting and spalling, releasing iron debris and causing torque ripple and vibration impacts.
- **Wear**: the sliding component gradually removes material, clearances grow and preload changes.
- **Fretting wear**: in small-angle reciprocating motion the contact zone is not fully rolled over and grease cannot flow back, producing false brinelling and local wear.

Wear usually does not proceed at a constant rate. Figure 1.1 shows the common three-stage picture: wear is fast during running-in, then enters a long steady stage, and finally an accelerating stage once surface damage spreads. For predictive maintenance the key is **to detect the change in slope during the steady stage and to schedule replacement before the turning point**; past the turning point, the remaining life extrapolated with a straight line is seriously overestimated (Section 3.7, Section 4.5).

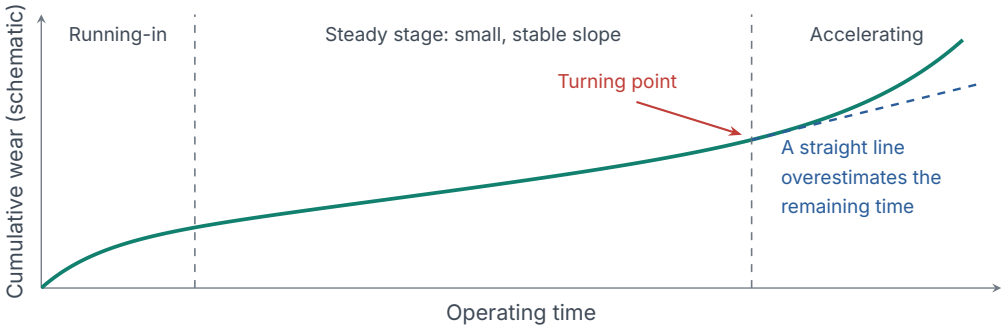


Figure 1.1 The three stages of wear (schematic). The blue dashed line is a straight-line extrapolation at the turning point and is clearly slower than reality.

1.2 The power law of fatigue life

The basic rating life of rolling bearings (ISO 281, derived from the theory of Lundberg and Palmgren) is

$$L_{10} = \left(\frac{C}{P}\right)^{p} \quad [\text{million revolutions}], \qquad p = \begin{cases} 3 & \text{ball bearings} \\ 10/3 & \text{roller bearings} \end{cases} \tag{1.1}$$

where C is the basic dynamic load rating, P is the equivalent dynamic load, and L_{10} is the number of revolutions that 90% of bearings reach. The exponent p comes from contact-fatigue tests: if the load doubles, life falls to $1/2^{p}$, or about $1/10$.

RV reducer catalogs give life in the same form:

$$L_h = L_0 \cdot \frac{N_0}{N_m} \cdot \left(\frac{T_0}{T_m}\right)^{10/3} \tag{1.2}$$

L_0 is the rated life (hours), T_0 the rated torque, N_0 the rated output speed, and T_m and N_m the equivalent torque and average speed in actual operation. The exponent is $10/3$ because the critical components inside the reducer are roller and needle bearings. Converting hours into revolutions, the total number of revolutions that can be sustained under rated conditions is

$$R_0 = 60\, L_0\, N_0 \quad [\text{revolutions}] \tag{1.3}$$

For example, with the commonly cataloged $L_0 = 6000\,\text{h}$ and $N_0 = 15\,\text{rpm}$, $R_0 = 60 \times 6000 \times 15 = 5{,}400{,}000$ revolutions. The actual T_0, N_0 and L_0 depend on the model; use the catalog of the reducer in use.

1.3 Variable load: Miner's linear accumulation and equivalent torque

Robot torque changes sharply from one motion to the next, so the variable load has to be converted into one equivalent value. Miner's rule assumes that the fractions of life consumed by each load segment can be added linearly: if n_i revolutions are made at torque T_i, and the life at that torque is N_i revolutions, then

$$\text{consumed fraction} = \sum_i \frac{n_i}{N_i}, \qquad \text{rated life is reached when it reaches 1.} \tag{1.4}$$

The "life" here is the rated life L_{10}: of identical components under the same conditions, about 90% are still running and about 10% have failed. A consumed fraction of 1 therefore means that the statistical high-risk zone has been entered, not that this particular unit must fail at that moment; the actual life of individual components is calibrated with Weibull statistics and replacement records (Section 5).

Derivation: equivalent torque

From Eq. (1.2), life expressed in revolutions is $N_i = R_0\,(T_0/T_i)^p$. Substituting into Eq. (1.4):

$$\sum_i \frac{n_i}{N_i} = \frac{1}{R_0}\sum_i n_i \left(\frac{T_i}{T_0}\right)^p .$$

The equivalent torque T_m is defined as the constant torque that "consumes the same life over the same total number of revolutions $\sum n_i$":

$$\frac{1}{R_0}\Big(\sum_i n_i\Big)\left(\frac{T_m}{T_0}\right)^p = \frac{1}{R_0}\sum_i n_i \left(\frac{T_i}{T_0}\right)^p \quad\Longrightarrow\quad T_m = \left(\frac{\sum_i n_i\,|T_i|^p}{\sum_i n_i}\right)^{1/p} .$$

This is exactly the catalog formula for equivalent torque weighted by "time × speed" ($n_i = t_i N_i^{\mathrm{rpm}}$).

Replacing the discrete segments with continuous time, the revolutions n_i become the angle increment $|\omega|\,\mathrm{d}t/2\pi$, which gives the basic formula with which DTM calculates damage:

$$\boxed{D = \frac{1}{2\pi}\int \left(\frac{|T(t)|}{T_0}\right)^p |\omega(t)|\,\mathrm{d}t = \frac{1}{2\pi}\int \left(\frac{|T|}{T_0}\right)^p |\mathrm{d}\theta|} \qquad \text{consumed fraction} = \frac{D}{R_0} \tag{1.5}$$

The unit of D is "equivalent revolutions at rated torque": one revolution at rated torque increases D by 1. The second equality in Eq. (1.5) rewrites the time integral as an integral over angle; this form is used in Section 7. It says that **whether the same path is travelled quickly or slowly, the damage is the same as long as the torque does not change**.

1.4 Sensitivity: 10% more torque, 27% less life

Taking the logarithmic derivative of $L \propto T_m^{-p}$:

$$\frac{\mathrm{d}L}{L} = -p\,\frac{\mathrm{d}T_m}{T_m} \qquad (p = 10/3) \tag{1.6}$$

Each 1% increase in equivalent torque reduces life by about 3.3%. For finite changes the ratio is calculated directly; see Table 1.1 and Figure 1.2.

Table 1.1 Effect of a change in equivalent torque on RV reducer life ($p = 10/3$)

Change in equivalent torque	−20%	−10%	+5%	+10%	+20%	+30%
Life multiplier $(T_m'/T_m)^{-10/3}$	2.10	1.42	0.85	0.73	0.54	0.42

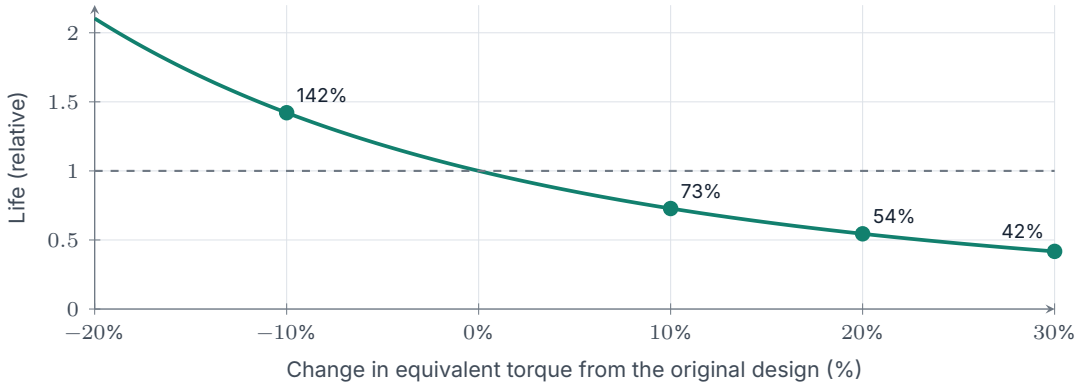


Figure 1.2 Life versus equivalent torque. Raising acceleration to shorten cycle time increases inertial torque, and life falls with the $10/3$ power.

Illustrative calculation: how different can life be for the same program?

Suppose an axis turns a total of 240° ($2/3$ revolution) back and forth per cycle, at 10 cycles per minute, running around the clock: $14{,}400$ cycles and $9{,}600$ revolutions per day. With $R_0 = 5{,}400{,}000$ revolutions:

- $T_m = T_0$: $5{,}400{,}000/9{,}600 \approx 563$ days (about 1.5 years).
- $T_m = 1.2\,T_0$ (higher acceleration): $563 \times 0.545 \approx 306$ days.
- $T_m = 0.9\,T_0$ (slowing motions that do not affect cycle time): $563 \times 1.42 \approx 800$ days.

Every single motion may be within the allowable range; the difference comes entirely from accumulation. This is the problem addressed in Section 6 and Section 7.

1.5 Single-event checks and cumulative damage

Catalogs also specify two "single-event" limits: the allowable torque at start and stop (taking a common RV catalog as an example, about 2.5 times rated for most models and about 2 times for the smallest model) and the momentary maximum allowable torque (about 5 times, with a limited number of occurrences). Actual values depend on the model; use the catalog of the model in use. These two limits only prevent damage from **a single overload**; they do not deal with cumulative fatigue.

Heavy-load handling programs often accelerate and stop abruptly 5–10 times a minute: $7{,}200$–$14{,}400$ times a day, or 216,000–432,000 times a month. Even if every event stays below the start/stop allowable torque, damage accumulated with the $10/3$ power can far exceed the value estimated from the average load at selection. DTM therefore calculates two things at the same time: the ratio of the peak to the allowable value and the number of exceedances (single-event check), and the cumulative damage of Eq. (1.5) (Section 6.5).

1.6 Load concentration and fretting wear

Eq. (1.1) and Eq. (1.2) implicitly assume that the load is spread evenly over the whole raceway and all rolling elements. Robots often do the opposite:

- **The same angle range is loaded repeatedly.** When the same pick-and-place motion is repeated tens of thousands of times a day, the load concentrates on the same range of output angles of the reducer, that is, on the same range of tooth flanks, pins and bearing raceways. The local equivalent load is higher than the average and the local life is shorter than the overall estimate.
- **Small-angle reciprocation.** When the swing amplitude is smaller than the width of the rolling contact, the rolling elements do not fully roll over the contact zone and grease is not easily carried back to the contact surface; even with modest average torque, false brinelling and fretting wear occur.

Neither of these can be seen from the "average torque"; torque has to be mapped to **angle**. DTM Pro quantifies the concentration with an angle concentration map and compares the friction of frequently and rarely used angle regions with a full-range sweep (Section 6.6, Section 6.7).

1.7 Temperature and grease aging

Grease deterioration (oxidation of the base oil, breakdown of the thickener structure) is a chemical reaction whose rate increases with temperature in the Arrhenius form:

$$k_r(T) = A\,\exp\!\left(-\frac{E_a}{R\,T_K}\right) \tag{1.7}$$

In the usual temperature range this corresponds to the rule of thumb that "the reaction rate increases 2–3 times for about every 10 °C rise" (van 't Hoff's rule). Grease life is often estimated to "halve for every 10–15 °C rise"; DTM takes the more conservative "doubles every 10 °C". Grease-change intervals specified in hours should therefore be shortened proportionally at high temperature:

$$h_{\text{eff}} = \int 2^{\max(0,\,(T(t)-T_{\text{ref}})/10)}\,\mathrm{d}t \tag{1.8}$$

h_{eff} is the "equivalent aging hours" and T_{ref} the reference temperature. Aging is accelerated only above the reference temperature and is not extended at low temperature, which is the conservative choice.

DTM implementation

- Damage is calculated in the angle form of Eq. (1.5): $D = \sum |\Delta\theta|/360^\circ \cdot (|T|/T_0)^p$; $p = 10/3$ for RV reducers and racks and $p = 3$ for gearboxes, settable in the configuration file.
- If the start/stop allowable torque and the momentary maximum allowable torque are not entered, 2.5 and 5 times the rated torque are used as defaults for RV reducers; these are only defaults, smaller models may be lower, and the catalog values of the model in use should be entered.
- Effective grease aging: reference temperature $40\,^\circ$C by default, aging rate doubles for every $10\,^\circ$C above it; the grease-change interval is whichever comes first of hours or years.

2 Measurement principle: obtaining friction from disturbance torque

DTM does not add a torque sensor; it uses the "disturbance torque" that the controller already computes. This chapter explains what disturbance torque contains physically, how friction is separated from it by subtracting forward and reverse runs, and what conditions this approach needs.

2.1 Equations of motion of a robot joint

The joint torques of an n-axis serial robot satisfy (Lagrange form):

$$\tau = \mathbf{M}(\mathbf{q})\,\ddot{\mathbf{q}} + \mathbf{C}(\mathbf{q},\dot{\mathbf{q}})\,\dot{\mathbf{q}} + \mathbf{g}(\mathbf{q}) + \tau_f(\dot{\mathbf{q}},\, T,\, \text{condition}) + \tau_{\text{ext}} \tag{2.1}$$

- $\mathbf{M}(\mathbf{q})\ddot{\mathbf{q}}$: inertia term, proportional to angular acceleration, including the mass distribution of the arm and the workpiece;
- $\mathbf{C}(\mathbf{q},\dot{\mathbf{q}})\dot{\mathbf{q}}$: Coriolis and centrifugal forces, proportional to products of velocities;
- $\mathbf{g}(\mathbf{q})$: gravity term, depending only on posture (including the balancer);
- τ_f: friction of the reducer, bearings and oil seals, depending on speed, temperature and **wear condition**; this is the quantity to be monitored;
- τ_{ext}: external force (collision, workpiece interference).

Motor-side torque is amplified to the joint by the reduction ratio N and efficiency η_g: $\tau_j \approx N\,\eta_g\,K_t\,i$, where K_t is the motor torque constant and i the current.

2.2 What disturbance torque contains

For collision detection and feedforward control, the controller uses its built-in dynamic model (including the payload set by the user) to calculate "how much torque the current motion should need", and compares it with the actual torque derived from the current. The difference is the disturbance torque:

$$\tau_{\text{dis}} = \tau_{\text{act}} - \underbrace{\left(\hat{M}\ddot{q} + \hat{C}\dot{q} + \hat{g}\right)}_{\text{model torque}} \;\approx\; \tau_f + \underbrace{\Delta M\,\ddot{q} + \Delta g}_{\text{model and payload-setting error}} + \tau_{\text{ext}} + \varepsilon \tag{2.2}$$

ΔM and Δg are the differences between model and reality (inaccurate payload setting, aging balancer), and ε is measurement noise. The actual algorithms of each manufacturer are not published; DTM relies only on the general structure of Eq. (2.2). This is an **assumption**: whether the values actually read from the controller fit this structure (unit, scale, filtering) must be confirmed at commissioning (Section 9.3). Two important consequences follow:

(1) In **constant-speed** ($\ddot{q} = 0$) motion **without external force**, what mainly remains in the disturbance torque is friction τ_f and the gravity-model error Δg.

(2) Disturbance torque does **not** contain the gravity and inertia already removed by the model. It is therefore suitable for monitoring friction but cannot be used directly to calculate the cumulative damage of the reducer: damage requires the total torque actually transmitted by the reducer (Section 6.5).

2.3 Friction model of the reducer

The variation of friction with relative speed can be described by the Stribeck model:

$$\tau_f(\omega) = \left[F_c + (F_s - F_c)\, e^{-(|\omega|/\omega_s)^{\delta}}\right] \operatorname{sgn}(\omega) + F_v\, \omega \tag{2.3}$$

F_s is static friction, F_c Coulomb (constant) friction, F_v the viscous friction coefficient, and ω_s the Stribeck speed. When the speed is much greater than ω_s, the exponential term vanishes, leaving

$$\tau_f(\omega) \approx F_c\, \operatorname{sgn}(\omega) + F_v\, \omega \tag{2.4}$$

The two parameters correspond to different physics:

- F_v comes mainly from the shear resistance of the lubricant and is proportional to viscosity. Grease viscosity falls roughly exponentially with temperature, so F_v is temperature-sensitive; grease deterioration, overfilling and low temperature all raise F_v.
- F_c comes mainly from solid contact and boundary lubrication. It rises when surface roughening, pitting or wear change the preload and contact condition; it is less sensitive to temperature.

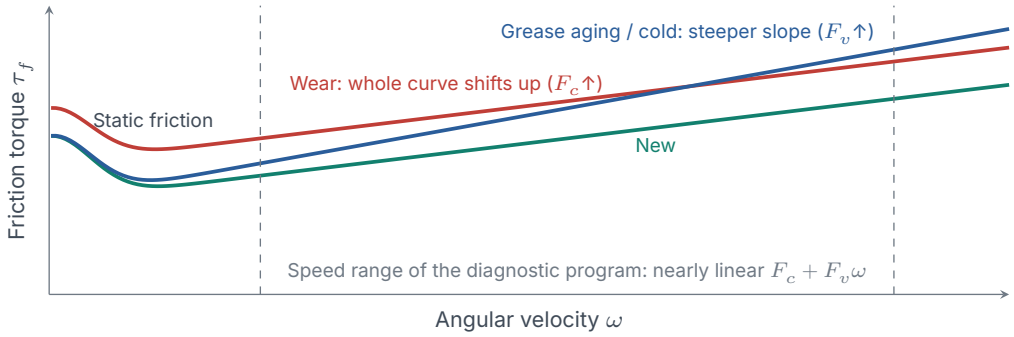


Figure 2.1 Friction versus speed (schematic). In the speed range used by the diagnostic program, friction is nearly linear; wear mainly raises the intercept, while grease and temperature mainly change the slope.

Temperature dependence: the viscosity of lubricants is often approximated exponentially (Reynolds viscosity model), $\mu(T) = \mu_{\text{ref}}\, e^{-b\,(T - T_{\text{ref}})}$. Since $F_v \propto \mu$,

$$\ln F_v = a - k_v\, T, \tag{2.5}$$

that is, for each 1 °C rise in temperature F_v falls by roughly the fraction k_v. This is the basis of the temperature compensation in Section 3.4.

2.4 Subtracting forward and reverse runs to separate friction

The diagnostic program moves a single joint once forward and once in reverse over the same angle range at a constant speed v. Consider the same position θ:

Derivation: F and G

At constant speed $\ddot{q} = 0$ and the inertia term vanishes. When only one joint moves, that joint's own velocity-product term is also zero: the inertia M_{ii} of joint i does not change with its own

angle q_i ($\partial M_{ii}/\partial q_i = 0$), so $[\mathbf{C}\dot{\mathbf{q}}]_i = 0$ when only $\dot{q}_i$ is non-zero. From Eq. (2.2) and Eq. (2.4):

$$\tau_+(\theta) = \Delta g(\theta) + (F_c + F_v v) + \varepsilon_+, \qquad \tau_-(\theta) = \Delta g(\theta) - (F_c + F_v v) + \varepsilon_-.$$

The gravity error Δg depends only on posture and is the same in both directions; friction changes sign with direction. Therefore

$$F(v) = \frac{\tau_+ - \tau_-}{2} = F_c + F_v v + \frac{\varepsilon_+ - \varepsilon_-}{2}, \qquad G = \frac{\tau_+ + \tau_-}{2} = \Delta g + \frac{\varepsilon_+ + \varepsilon_-}{2}. \tag{2.6}$$

F is friction and G is the direction-independent offset (payload setting, balancer, gravity-model error). A change in G indicates "a setting or balancer problem, not the reducer", which prevents an incorrect payload setting from being misjudged as wear.

Conditions: (1) the forward and reverse runs cover the same angle range so that Δg is the same; (2) only the constant-speed part is used, cutting off the acceleration and deceleration at both ends; (3) the speed override is 100% so that v is the same; (4) the runs are made after warm-up with the same payload, so that temperature and Δg are comparable.

Noise and number of samples: if the noise of each sample is independent with standard deviation σ_ε and n samples are averaged per segment, then

$$\mathrm{SD}(F) = \frac{1}{2}\sqrt{\frac{\sigma_\varepsilon^2}{n} + \frac{\sigma_\varepsilon^2}{n}} = \frac{\sigma_\varepsilon}{\sqrt{2n}}. \tag{2.7}$$

With a 2-second constant-speed segment at 5 samples per second, $n = 10$ and the noise of F is about $1/\sqrt{20} \approx 0.22$ times that of a single sample. If the noise is correlated in time the actual improvement is smaller, so DTM additionally absorbs the remaining variation with baseline statistics over many diagnostic runs (Section 3).

Averaging over the angle range: $\Delta g(\theta)$ actually varies with angle. As long as the forward and reverse runs pass through the same set of angles, the average of Δg still cancels when $\overline{\tau_+} - \overline{\tau_-}$ is computed from whole-segment averages. Point-by-point alignment of angles is therefore not needed, but the start and end angles of the forward and reverse runs must be the same.

2.5 Multi-speed regression and ripple

Fitting a least-squares straight line to $F(v_j)$ obtained at m speeds $v_1, \ldots, v_m$ (for example 5%, 10%, 20%) gives, from Eq. (2.4),

$$\hat{F}_v = \frac{\sum_j (v_j - \bar{v})(F_j - \bar{F})}{\sum_j (v_j - \bar{v})^2}, \qquad \hat{F}_c = \bar{F} - \hat{F}_v\,\bar{v}. \tag{2.8}$$

With three speeds there is only one degree of freedom for estimating the residual, so DTM uses "F at the reference speed (10%)" as the main representative value, and F_c and F_v to distinguish failure types.

Ripple: the standard deviation of disturbance torque within the constant-speed segments,

$$\text{ripple} = \frac{1}{2m}\sum_j \Big[\mathrm{SD}(\tau_{+,j}) + \mathrm{SD}(\tau_{-,j})\Big]. \tag{2.9}$$

A healthy reducer has smooth torque in the constant-speed segment; tooth pitting, spalling or bearing damage cause torque fluctuations that recur periodically with angle. At 2–5 samples per second, the samples in a constant-speed segment fall at different angles, so the scatter between samples grows with local damage. This is a time-domain statistic that needs neither high-speed sampling nor spectrum analysis.

2.6 Features and failure types

Table 2.1 Feature changes and possible causes (example rules, to be corrected according to teardown results)

Observed change	Possible cause and reason
Ripple $\geq 1.4\times$ and $F \geq 1.2\times$	Reducer wear or local damage: uneven contact causes both higher friction and fluctuation over angle
Ripple $\geq 1.8\times$ (when F is below $1.2\times$)	Suspected spalling or pitting: a local defect causes large, periodic torque changes
Mainly F_v rises ($\geq 1.15\times$, and more than 1.1 times the F_c ratio)	Grease deterioration, overfilling or low temperature: higher viscous resistance
Mainly F_c rises	Reducer wear, brake drag or abnormal preload: more solid contact
G shifts	Balancer or payload setting: independent of direction, not friction

2.7 How high a sampling rate is needed

The required sampling rate is determined by the phenomenon to be observed; higher is not automatically better:

- **Continuous monitoring** (2–5 samples per second): what is needed are the mean and scatter of constant-speed segments and the daily trend. The precision of the mean is determined by the number of samples (Eq. (2.7)), not by the sampling rate, so reading registers through the controller's web server is sufficient.
- **Program-level diagnosis** (50–125 samples per second): the 0.2–1-second acceleration and deceleration must be resolved clearly and mapped to program lines. By the sampling theorem the sampling rate must be at least twice the highest frequency component of the signal; the frequency content of acceleration and deceleration is mostly from a few Hz to a little over ten Hz, so 50 or more samples per second is sufficient.
- **Impacts and gear spectra** (several thousand samples per second or more): the impacts of emergency stops and collisions and the mesh frequencies are on the millisecond scale and need an accelerometer; polling controller registers cannot support spectrum analysis.

DTM implementation

- Diagnostic program: J2 and J3 each run forward and reverse at 5%, 10% and 20% speed; the segment code $=$ axis $\times 1000+$ speed% $\times 10+$ direction is written to `R[110]`; segments run at a speed override other than 100% are discarded.
- 1.0 second is cut from each end of a segment; a segment is used only if the constant-speed part is at least 2.0 seconds long and has at least 3 samples.
- F at the 10% reference speed is the main representative value; if that speed is missing it is calculated from the regression of Eq. (2.8). $s_{\text{dir}} = \pm 1$ is set according to the mounting direction of the axis so that F is positive.
- Travel axes (horizontal linear) have no gravity moment, so G is not evaluated.

3 Detection: baselines, statistics and false-alarm control

Once F, F_c, F_v and ripple are available from each diagnostic run, the next step is to decide whether "this value is abnormal". DTM's principle is that **each axis is compared only with itself**, and a series of steps first removes changes that are not wear (temperature, season, factors that change all axes together, occasional noise) before deciding.

3.1 Why use the axis's own baseline

Even for the same axis of robots of the same model, the absolute value of friction differs markedly with assembly, preload, amount of grease and payload; the unit and scale of disturbance torque also depend on controller settings. Judging with absolute thresholds would require calibration with a large amount of data from the same model, and would still not transfer to another unit. Comparing with the axis's own baseline has three advantages:

- **Dimensionless**: the ratio $r = x/\mu$ does not depend on unit or scale, so no calibration to N m is needed first.
- **Absorbs individual differences automatically**: the "normal" of each axis is defined by its own first N runs.
- **Proportional to the physical quantity**: doubled friction means $r = 2$, whatever the original value.

$$\mu = \frac{1}{N}\sum_{i=1}^{N} x_i, \qquad s = \sqrt{\frac{1}{N-1}\sum_{i=1}^{N}(x_i-\mu)^2}, \qquad r = \frac{x}{\mu} \tag{3.1}$$

x_i are the first N valid diagnostic runs after a reset (new unit, grease change, replacement), after temperature compensation (Section 3.4). Grease changes and replacements cut the data into independent periods, and each period builds its own baseline. In addition, a cumulative ratio "relative to new" is kept that is not reset by a grease change, so that ongoing wear is not "washed out" (Section 4.8).

3.2 z-scores and a floor on the standard deviation

The ratio is the main criterion; the z-score is used only for **earlier** detection:

$$z = \frac{x-\mu}{\max(s,\ 0.05\,|\mu|)} \tag{3.2}$$

Why a floor is needed: estimating the standard deviation from only $N = 10$ values is itself imprecise. For a normal population, the relative standard error of the sample standard deviation is about

$$\frac{\mathrm{SE}(s)}{\sigma} \approx \frac{1}{\sqrt{2(N-1)}} = \frac{1}{\sqrt{18}} \approx 0.24 \qquad (N = 10). \tag{3.3}$$

If the baseline period happens to be very stable, s can be underestimated by half or more, the z-score is inflated, and any small change becomes "3 standard deviations". A floor of $0.05|\mu|$ (5% of the baseline) prevents such false alarms.

Decision rules (example values):

- $r \geq 1.5$ → caution; $r \geq 2.0$ → alert (main criterion).
- $z \geq 2$ and $r \geq 1.15$ → raised early to caution (a small change, but clearly beyond the usual fluctuation).
- Already at the caution ratio and $z \geq 3$ → raised to alert.

The z conditions also require a minimum ratio so that, on a very stable axis, normal fluctuations within 5% are not judged abnormal.

3.3 Hysteresis: consecutive confirmation reduces false alarms

A single diagnostic run can be affected by chance factors (a cold start right after a stop, interference during handling, momentary noise when reading). DTM raises the level only when **2 consecutive** runs meet the condition, and lowers it only after 2 consecutive runs below it.

If the false-alarm probability of a single run is p and the runs are independent, the probability of k consecutive false alarms is p^k. With $p = 0.05$ and $k = 2$ the false-alarm rate falls to 0.0025, one twentieth of the original. In practice two consecutive runs are often affected by the same conditions (the same cold spell, the same batch of products) and are positively correlated, so the real reduction is less than $1/20$; this is also why temperature compensation and cross-axis removal are done first. The cost is a confirmation delay of one diagnostic run (one day with one run per day), which has little effect on wear trends that develop over weeks.

3.4 Temperature compensation

By Eq. (2.5), friction varies approximately exponentially with temperature. Each value is converted to the mean temperature of the baseline period, T_{ref}:

$$x_{\text{comp}} = x \cdot e^{\,k\,(T - T_{\text{ref}})} \tag{3.4}$$

When the temperature is above the reference, raw friction is lower and is restored by multiplying by a factor greater than 1, and vice versa.

Where the coefficients come from: (1) Defaults: about 0.025/°C for F_v (viscous friction rises about 2.5% for every 1 °C drop) and about 0.004/°C for F_c; for F the two are weighted by their shares at the reference speed:

$$k_F = \frac{k_c\,F_c + k_v\,F_v\,v_{\text{ref}}}{F} \tag{3.5}$$

(2) Estimated from data: when slow deterioration is going on at the same time, regressing on temperature alone mistakes deterioration for a temperature effect, so a regression with two explanatory variables is used:

$$\ln x_i = a - k\,T_i + c\,d_i + e_i \tag{3.6}$$

d_i is the day number and $c\,d_i$ absorbs the slow trend. Least squares gives k and its standard error; the estimate is not used if the temperature range is too narrow (for example below 8 °C) or there are too few samples.

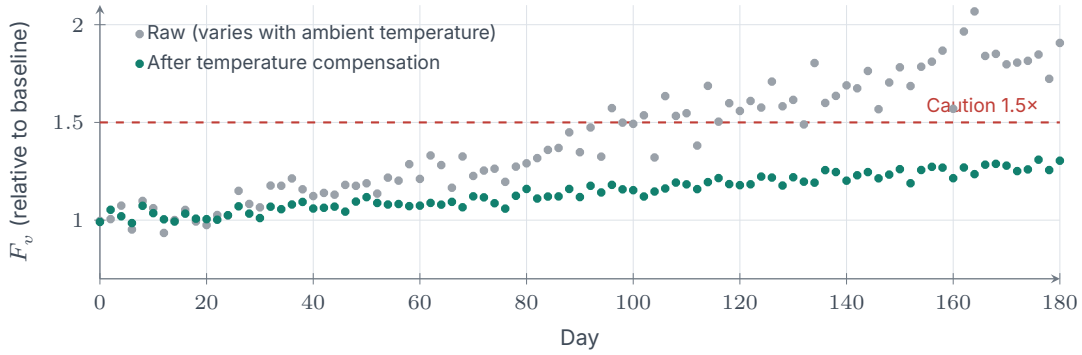


Figure 3.1 Effect of temperature compensation (simulated data, $k_v = 0.025$/°C). Raw values move with the seasons and with day–night temperature differences and may be misjudged at the lowest temperatures; after compensation only the slow deterioration trend remains.

3.5 Removing common-mode changes across axes

Even without a temperature sensor, ambient temperature, cold starts and changes in production cycle make **the friction of all axes change together**; when a single reducer wears, only that axis changes. The change of each axis is decomposed as

$$\ln x_j(t) = \ln x_j(0) + c(t) + d_j(t) + e_j(t) \tag{3.7}$$

where $c(t)$ is the change common to all axes and $d_j(t)$ the change specific to axis j (for example wear).

Method: estimating the common change with the median

1. For each axis, accumulate the successive log-changes of **viscous friction** F_v into a chained index $L_j(t) = \sum \ln(F_{v,j}(t_i)/F_{v,j}(t_{i-1}))$; steps across a grease change or replacement are not included. If temperature is available, temperature compensation is applied first.
2. Common change $\hat{c}(t) = \text{median}_j\, L_j(t)$. The median is used rather than the mean: as long as fewer than half of the axes are deteriorating, the median lies on a healthy axis and is not pulled away by a single deteriorating axis. With three or more axes the axis itself is included in the median; with only two axes only the other axis can be used.
3. $\hat{c}$ is converted into an equivalent temperature difference $\Delta T_{\text{eq}} = -\hat{c}/k_v$ (limited to ± 30 °C), which is converted into a factor $m_k = e^{-k_k \Delta T_{\text{eq}}}$ with each indicator's own temperature coefficient and removed as x/m_k.

Why the common change is taken only from F_v: temperature, cold starts and cycle conditions mainly affect the viscous term. If the median were also taken of F_c, then when two axes wear at the same time (for example reducers replaced in the same batch), the common rise in F_c would be removed as an environmental change. Converting the common change first into an "equivalent temperature" and then applying the very small temperature coefficient of F_c leaves the wear signal in F_c almost untouched.

The price: deterioration of all axes at the same rate at the same time (for example grease from the same batch aging together) is removed. That case is guarded separately by grease life and iron

debris (Section 4). When a temperature signal is available, it is possible to configure temperature compensation only, without removing the common change.

3.6 Monitoring during production: robust statistics

If you do not want to stop production for the diagnostic program, "motion codes" can be written before and after fixed motions in the production program, and DTM records the mean, RMS, peak and standard deviation of each axis's disturbance torque for that motion in every cycle. Production motions include acceleration, deceleration and load changes, so the scatter of individual cycles is large; the **daily median** is used instead:

$$\tilde{x}_{\text{day}} = \text{median}\{x_1, \dots, x_n\}, \qquad n \geq 20 \text{ valid cycles} \tag{3.8}$$

The breakdown point of the median is 50%: even if some cycles are affected by interference, pauses or changes of the speed override, the result is not pulled. The comparison is with the baseline of the first 10 valid days after a reset.

Friction is only part of production torque; a 50% rise in friction may show up as only 10–20% in total torque, so the thresholds are lower: caution at 1.15 times, alert at 1.30 times, confirmed on 2 consecutive valid days. If the taught points or speed of the motion are changed, the baseline of that code must be reset.

3.7 Trend extrapolation and detection of acceleration

The most recent n compensated values y_i (for example 30) are fitted by least squares to a straight line $y = a + b\,x$ against time x_i (days from today, $x \leq 0$); a is today's estimate. The number of days until threshold y^* is reached is

$$t^* = \frac{y^* - a}{b} \qquad (b > 0;\ 0 \text{ if } a \geq y^*) \tag{3.9}$$

In the late stage wear often speeds up (Figure 1.1), and a straight-line extrapolation overestimates the remaining time. DTM detects acceleration with two methods and judges it as accelerating if either holds:

(a) **Ratio of first-half and second-half slopes**: the window is split into two halves, each regressed separately; the second-half slope $b_2 > 0$ and $b_2 \geq 1.5\,b_1$.

(b) **Quadratic regression**: $y = \alpha x^2 + \beta x + \gamma$ with curvature $\alpha > 0$, slope at the latest point $\beta \geq 1.5\,b$, and a residual sum of squares of the quadratic no more than 0.8 times that of the line. The last condition keeps noise from being judged as acceleration: one more parameter always reduces the residuals, so only a large enough reduction counts as real curvature.

When acceleration is detected, the **shortest** of three estimates is used: the line over the whole window, the line over the second half, and the intersection of the quadratic with the threshold,

$$t^*_{\text{quad}} = \frac{-\beta + \sqrt{\beta^2 - 4\alpha(\gamma - y^*)}}{2\alpha}. \tag{3.10}$$

Uncertainty: extrapolated days vary widely with noise. DTM shows an empirical range of $\pm(30\% \times t^* + 7 \text{ days})$ and does not show values beyond 730 days. A rigorous statistical prediction interval

requires independent residuals and a correct linear model, and wear trends usually satisfy neither, so a conservative empirical range is used; its width should be corrected with field data.

DTM implementation

- Baseline $N = 10$; ratio thresholds 1.5/2.0; z thresholds 2/3, early escalation requires a ratio ≥ 1.15; standard-deviation floor 5%; 2 consecutive confirmations.
- When only one axis is monitored and the temperature spread of the baseline period is ≥ 5 °C, the temperature coefficient is regressed from the baseline data; when several axes are monitored, the default coefficients are always used so that the common-change indices of all axes are consistent.
- Common change: used by default when two or more axes are monitored; when set to `"auto"` it is used only when there is no temperature signal.
- Remaining days are the minimum of the extrapolations from torque, production cycles, iron debris and vibration envelope; a friction rise judged to be caused by the motor is not included.

4 Multi-evidence fusion and maintenance decisions

Rising friction does not necessarily mean the reducer is failing: grease deterioration, a wrong payload setting, brake drag and motor-bearing damage all raise disturbance torque. Replacing a reducer is costly in money and downtime, and so is a wrong decision. This chapter explains how DTM combines torque, iron debris, vibration, temperature and component life into three decisions: change the grease, replace the reducer, inspect the servo motor.

4.1 Evidence families

DTM groups evidence into families that are **physically independent** of each other: a given measurement error or non-wear factor usually affects only one of the families.

Table 4.1 Evidence families and their components (coefficients are example values, adjustable in the configuration file)

Family	Components (maximum is taken)	Physical source
Torque — wear type	F_c, ripple, $0.8 \times F$, cumulative F_c and ripple relative to new, $0.8\times$ production cycle	Solid contact and surface damage
Torque — viscous type	F_v	Lubricant viscosity
Iron debris	Concentration, generation rate W	Material removal (mass)
Vibration — reducer type	Radial envelope, mesh, kurtosis, $0.8\times$ axial RMS, $0.8\times$ axial ratio, production-cycle vibration	Contact impacts and gear mesh
Vibration — motor type	Motor rotation frequency (larger of radial and axial)	Motor bearings and rotor
Temperature	Sudden rise of housing temperature rise (weight 0.6)	Frictional heat

4.2 Severity function

Indicators have different units and thresholds, so they are first converted into a common dimensionless "severity" $s \in [0,\ 1.5]$: 0.5 is caution and 1.0 is alert. For ratio indicators (thresholds: caution c, alert a):

$$s(r) = \begin{cases} 0 & r \le 1 \\ 0.5\,\dfrac{r-1}{c-1} & 1 < r \le c \\ \min\!\left(1.5,\ 0.5 + 0.5\,\dfrac{r-c}{a-c}\right) & r > c \end{cases} \tag{4.1}$$

For absolute indicators (for example iron-debris concentration) a floor value replaces 1. The advantage of the piecewise-linear form is that the meaning of the thresholds is kept ($s = 0.5$ is exactly the caution threshold) and that it is continuous near the thresholds, so a tiny difference does not jump a level. The upper limit of 1.5 prevents a single extreme value from dominating.

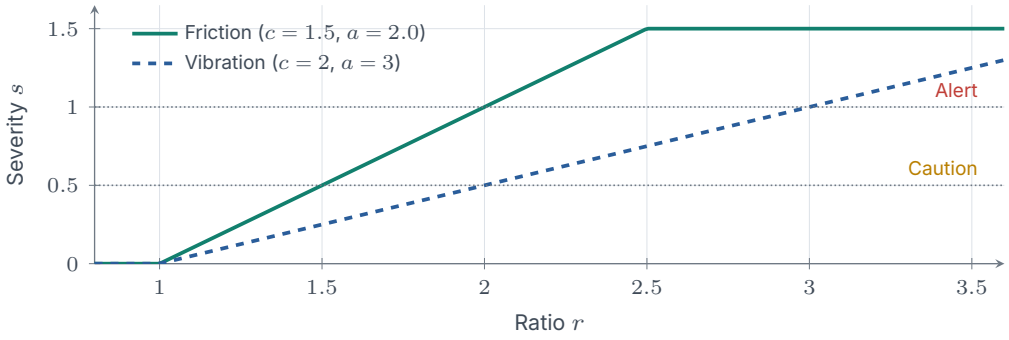


Figure 4.1 Severity function. Different indicators have different thresholds, but after conversion 0.5 is always caution and 1.0 alert.

4.3 A probabilistic reading of the two-evidence rule

Rule: when computing the score for "replace reducer" or "replace motor", the severities of the relevant families are sorted from largest to smallest. If only one family is ≥ 0.25, the score $= \min(s_{(1)}, 1.5) \times 0.45$; if two or more are, the score is the mean of the top two, $(s_{(1)} + s_{(2)})/2$.

Why two kinds of evidence are needed: a Bayesian view

Let H be "the reducer has deteriorated to the point of needing replacement" and E_k the observation of family k. If the families are mutually independent conditional on H, the posterior odds are the prior odds multiplied by the likelihood ratio of each piece of evidence:

$$\frac{P(H \mid E_1, E_2)}{P(\bar{H} \mid E_1, E_2)} = \frac{P(H)}{P(\bar{H})} \cdot \underbrace{\frac{P(E_1 \mid H)}{P(E_1 \mid \bar{H})}}_{\Lambda_1} \cdot \underbrace{\frac{P(E_2 \mid H)}{P(E_2 \mid \bar{H})}}_{\Lambda_2} \tag{4.2}$$

Numerical example: the prior odds that a reducer of this model needs replacement in a given period are $1:50$; the likelihood ratio of a friction rise is $\Lambda_1 = 10$ (common when there really is deterioration, but grease and setting problems cause it too).

- Torque evidence alone: odds $1:50 \times 10 = 1:5$, probability about 17%. Replacing on this basis would be wrong five times out of six.
- Adding a rise in iron debris ($\Lambda_2 = 10$): odds $1:50 \times 100 = 2:1$, probability about 67%.

Another angle is false alarms: if the probabilities that each family gives a false alarm for a non-wear reason are p_1 and p_2, the probability that both do so at the same time is about $p_1 p_2$, far smaller than either one.

DTM does not compute the likelihood ratios directly (estimating them needs a large amount of teardown data). It implements the same idea with "a single family goes no further than watch ($0.45\times$)": however strong one kind of evidence is, it only recommends "watch" or, in extreme cases, "prepare a spare", never replacement. Once teardown data accumulate (Section 8.4), the likelihood ratios of each family can be inferred from the actual detection and false-alarm rates and a true probabilistic model can be used.

4.4 Explaining away: motor or reducer

Rising friction has two main sources: the reducer, and the motor (bearings, brake drag). Their effects on disturbance torque are similar, but their effects on other evidence differ:

- Reducer deterioration: usually accompanied by rising iron debris or vibration at reducer frequencies (envelope, mesh).
- Motor bearing or brake: iron debris unchanged, vibration concentrated at the motor rotation frequency.

In a Bayesian network this is called "explaining away": when one cause holds, the probability of the other cause falls. DTM's rule: when friction rises but the severities of iron debris and reducer vibration are both below 0.5, while vibration at the motor rotation frequency rises, the torque-wear score is multiplied by

$$1 - 0.7\ \min(1,\ s_{\text{motor vibration}}) \tag{4.3}$$

and the finding is attributed to the motor. In a scenario of the built-in simulator, when the servo-motor bearing of the J4 gearbox deteriorates, the system reports "servo motor abnormality" rather than the gearbox.

4.5 Iron debris: mass balance and the turning point

Iron debris is the most direct evidence of reducer wear: the metal removed by wear stays in the grease.

Generation rate: with grease quantity M (g) and iron-debris concentration c (wt%), the mass of iron in the grease is $m_{\text{Fe}} = (c/100)\,M$. The generation rate between two samples (assuming constant grease quantity, uniform mixing and no loss) is

$$W = \frac{\Delta m_{\text{Fe}}}{\Delta h} = \frac{\Delta c}{100} \cdot \frac{M}{\Delta h} \cdot 1000 \quad [\text{mg/h}] \tag{4.4}$$

where Δh is the operating hours between the two samples. Concentration only reflects "how much is in the grease now" and returns to zero with a grease change; W reflects "how fast wear is progressing", is independent of grease changes and can be compared across grease batches.

Effect of sampling: when grease is drained at a grease change, the first grease out is old grease that had settled at the bottom or in the lines, with a higher concentration. By default DTM takes a "mid-stream" sample to reduce this bias.

Turning-point rule: once the concentration reaches the caution threshold, the remaining days are no longer extrapolated with a straight line; DTM shows "past the turning point, may deteriorate at any time" and recommends ordering a spare directly. If the generation rate accelerates twice in a row,

$$W_3 - W_2 > W_2 - W_1 > 0, \tag{4.5}$$

the level is raised by one. This corresponds to the accelerating stage of Figure 1.1: in that stage, any straight-line extrapolation underestimates how fast deterioration proceeds.

4.6 Vibration features

Vibration is measured by external sensors; DTM only reads feature values already computed by the sensor or gateway. The physics behind each feature:

- **RMS**: $\sqrt{\overline{x^2}}$, the vibration energy, reflecting the overall condition.
- **Envelope**: band-pass filtering followed by demodulation extracts the repetition rate of the impacts produced each time a local defect is rolled over. Local damage on bearings and tooth flanks raises the envelope before the RMS changes.
- **Kurtosis**: $\kappa = \mathrm{E}[(x-\mu)^4]/\sigma^4$. A normal distribution has $\kappa = 3$; when sharp impacts appear in the signal the tails become heavier and κ rises. Kurtosis is judged by absolute value (4 caution, 6 alert).
- **Mesh-frequency and motor-rotation-frequency energy**: correspond to the reducer gears and the motor rotor respectively and are used to tell the sources apart (Section 4.4).
- **Axial/radial ratio**: main-bearing damage or axial play increases axial vibration relative to radial.

4.7 Temperature: aging and sudden rises

Temperature has two completely different uses that must not be confused:

(1) **Steady high temperature**: accelerates grease aging (Eq. (1.8)); it only shortens the grease-change interval and is **not** treated as evidence of wear.

(2) **Sudden temperature rise**: under the same motion, a sudden increase in the difference between housing and ambient temperature, $\Delta T = T_{\text{housing}} - T_{\text{ambient}}$, indicates more frictional heat. Using the axis's own 14-day median as the baseline, a rise of 8 °C or more, or of 5 °C or more within a day, with no simultaneous rise in ambient temperature, is judged a "sudden temperature rise" caution.

The difference is used rather than the housing temperature to remove changes in weather and plant air conditioning; without an ambient sensor, the median housing temperature of the other axes on the same day and in the same motion serves as the reference (the same idea as the removal of common changes in Section 3.5). For 72 hours after a grease change the grease distribution is not yet stable and sudden rises are not judged. The temperature family has a weight of 0.6 and on its own reaches "watch" at most.

4.8 Three scores and decision thresholds

Table 4.2 The three scores for maintenance decisions (≥ 0.5: recommend grease change / prepare spare / inspect motor; ≥ 0.9: change grease now / schedule replacement)

Score	Calculation
Grease change	$\max(\text{torque viscous type},\ \text{grease-life severity},\ 0.8\min(\text{iron debris}, 1))$; grease life starts to score once 70% has been used
Reducer replacement	Two-evidence rule (torque wear type, iron debris, reducer-type vibration), or age risk if larger
Motor inspection	Two-evidence rule (motor vibration, friction rise without wear evidence), or age risk if larger

Recovery check after a grease change: after changing the grease and running for 2–4 weeks, compare the cumulative ratios of F_c and W relative to new. If they return to within 1.3 times, the cause was lubrication and has been resolved; if they remain high, the reducer itself has deteriorated and replacement should be prepared. This "intervene and watch the response" design uses the grease change as a diagnostic experiment that separates lubrication problems from mechanical damage.

Age risk: the Weibull life model (Section 5) gives the conditional failure probability within the spare-part lead time; at 0.2 DTM recommends preparing a spare, and at about 0.36 the score rises to 0.9 and replacement is recommended. This is preventive replacement based on life statistics; it is not limited by the two-evidence rule and is the only exception to "no replacement recommended on a single kind of evidence" in Section 9.5.

DTM implementation

- Family coefficients: F_c, ripple 1.0; F, production cycle 0.8; temperature rise 0.6; envelope, mesh, kurtosis 1.0; axial RMS, axial ratio 0.8; iron debris 1.0 (configuration section `[decision.weights]`).
- Single-family threshold 0.25, single-family coefficient 0.45; decision thresholds 0.5/0.9. If grease life is $\geq 100\%$ or iron debris is $\geq$ alert, the grease change is forced to "now"; when the reducer reaches "now", the grease change is merged into the replacement.
- Iron debris: starting thresholds for RV reducers 0.05/0.10 wt%; W at 2/3 times the as-new baseline; W is not calculated until 300 hours of operation after a grease change. The sampling interval shortens automatically to 60 days at caution and 21 days at alert.
- Vibration: ratios 2/3 times; kurtosis 4/6; axial/radial ratio 1.5 times baseline.

5 Life statistics: the Weibull distribution and spare parts

Monitoring answers "how is this unit doing now"; life statistics answer "how long does this kind of component usually last, and how many spares should be ready over the next few months". The two complement each other: even without a deterioration signal, age itself is a risk.

5.1 The Weibull distribution

The Weibull distribution is the most widely used life distribution in reliability engineering:

$$F(t) = 1 - \exp\left[-\left(\frac{t}{\eta}\right)^{\beta}\right], \qquad R(t) = 1 - F(t), \qquad h(t) = \frac{f(t)}{R(t)} = \frac{\beta}{\eta}\left(\frac{t}{\eta}\right)^{\beta-1} \tag{5.1}$$

η is the scale parameter (63.2% of components have failed by η) and β the shape parameter. The shape of the failure rate $h(t)$ maps directly to the failure mechanism (Figure 5.1):

- $\beta < 1$: failure rate falls with time; early failures (assembly, initial defects).
- $\beta = 1$: constant failure rate; random failures (exponential distribution).
- $\beta > 1$: failure rate rises with time; wear-out failures. Only in this case does "time-based preventive replacement" make sense; the larger β, the more concentrated the lives and the easier it is to schedule replacement.

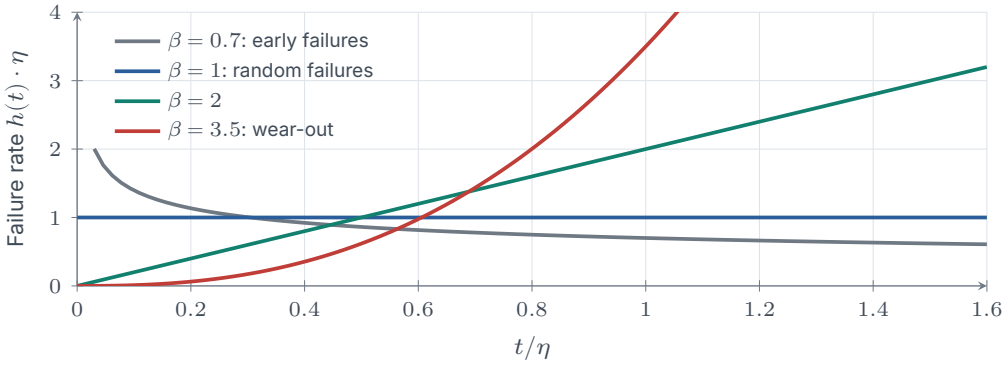


Figure 5.1 Weibull failure rate and shape parameter. For $\beta > 1$ the failure rate rises with service time, and age itself is a risk.

5.2 Maximum likelihood with censored data

Field life data come in two kinds: (1) **failures**: used until broken and then replaced, giving a complete life t_i; (2) **right-censored**: preventive replacement, no abnormality found at teardown, or still in service, so the life is only known to be "at least" t_j. Ignoring the second kind causes serious bias: using only the failures amounts to assuming that every component failed within the observation period, and life is underestimated.

Derivation: closed form for η and the profile likelihood

Let there be r failures and the rest censored. A failure contributes the probability density $f(t_i)$, a censored record the survival probability $R(t_j)$. The log-likelihood is

$$\ell(\beta,\eta) = \sum_{i\in\text{failures}} \left[\ln\beta - \beta\ln\eta + (\beta-1)\ln t_i\right] - \sum_{\text{all } k}\left(\frac{t_k}{\eta}\right)^{\beta}.$$

Differentiating with respect to η and setting to zero:

$$\frac{\partial\ell}{\partial\eta} = -\frac{r\beta}{\eta} + \frac{\beta}{\eta}\sum_{k}\left(\frac{t_k}{\eta}\right)^{\beta} = 0 \quad\Longrightarrow\quad \hat\eta(\beta) = \left(\frac{1}{r}\sum_{k} t_k^{\beta}\right)^{1/\beta}.$$

Substituting back leaves only one variable, β (the profile log-likelihood):

$$\ell_p(\beta) = r\ln\beta - r\beta\ln\hat\eta(\beta) + (\beta-1)\sum_{i\in\text{failures}}\ln t_i - r. \tag{5.2}$$

A grid search finds the β that maximizes ℓ_p, and η follows from the closed form. The one-dimensional search needs no starting value and does not diverge on small data sets as Newton's method can.

Numerical example: the effect of censored data

8 reducers of the same model: 3 failed at 18,000, 24,000 and 29,000 hours; 5 are still running normally at 30,000 hours.

	$\hat\beta$	$\hat\eta$ (h)	B_{10} (h)
3 failures only	6.21	25,533	17,771
Failures + 5 censored	4.36	35,628	21,263

Ignoring the 5 units still running underestimates η by about 28%, which makes preventive replacement too early and spare stock too large.

5.3 B_{10} life and conditional failure probability

B_{10} is the life at which 10% of components have failed:

$$B_{10} = \eta\,(-\ln 0.9)^{1/\beta}. \tag{5.3}$$

For a component that **has already run** t hours and is still normal, the probability of failing within the next Δ hours is a conditional probability:

$$P(\text{failure in } t\sim t+\Delta \mid \text{survival to } t) = 1 - \frac{R(t+\Delta)}{R(t)} = 1 - \exp\left[\left(\frac{t}{\eta}\right)^{\beta} - \left(\frac{t+\Delta}{\eta}\right)^{\beta}\right]. \tag{5.4}$$

This is the basis for deciding "whether to order a spare now": Δ is taken as the spare-part lead time.

Numerical example: spare-part lead time

$\beta = 3$, $\eta = 40{,}000$ h; an axis has run 30,000 h at 20 hours a day. The conditional probability

of failure within a 60-day spare lead time (1200 h) is 5.1%; within the next 12 months (about 7300 h) it is 32%. The fleet's "expected replacements in the next 12 months" is the sum of the conditional probabilities of all units.

5.4 Sparse data and fleet priors

A single plant often has few failure records, so the Weibull parameters are very uncertain. What DTM does:

- With fewer than 3 failure records, configured prior parameters are used and the source is shown on screen.
- The central server fits a "fleet prior" from the records of the same component type (for example "J2 reducer") at all plants and distributes it to each plant; edge hosts apply it automatically by version and mark it "from central". Pooling assumes that the life distributions of the same component type at different plants are similar; plants with very different operating conditions should be fitted separately.
- A replacement found "no abnormality" at teardown is changed to a censored record in the life statistics: the component did not actually fail and should not be counted as a failure.

DTM implementation

- β is grid-searched from 0.3 to 10 in steps of 0.01 and η uses the closed form; fitting starts with at least 2 failures, and priors take precedence below 3.
- Operating hours: interpolated or extrapolated from hour-meter readings when available, otherwise estimated from daily operating hours.
- Spares: if the conditional failure probability within the lead time (example 60 days) is ≥ 0.2, preparing a spare is recommended; if the remaining days are shorter than the lead time, immediate ordering is prompted; stock is allocated in order of each axis's remaining days and shortfalls are listed.

6 Program-level damage attribution (Pro)

Continuous monitoring answers "is the reducer getting worse"; Pro answers "**why**": which program line and which angle range are consuming life. During a short diagnostic period (3–7 days) the sampling rate is raised, each torque sample is mapped to the program line being executed and to the output angle of the reducer, and damage is accumulated per line and per angle with the damage formula of Section 1.3.

6.1 Problem setting

During the diagnostic period, 50–125 samples per second are acquired: the angles $\mathbf{q}(t)$ and torques $\mathbf{T}(t)$ of each axis and (optionally) the line number reported by the program. Three things have to be solved:

(1) **Alignment**: which program line each sample belongs to.

(2) **Attribution**: how much damage each line has accumulated, how high its peaks are and how long it dwells.

(3) **Spatial distribution**: which angle range of the reducer the damage is concentrated in.

6.2 Line-number alignment: why reported line numbers cannot be used directly

The most direct approach is to insert a line before each motion line that writes the line number to a register (the "instrumented program"). The problem lies in CNT (continuous, passing through points without stopping) motion: to make the path smooth, the controller executes the following non-motion instructions **ahead of time**, so while the robot is still on the path of the previous segment, the register already holds the number of the next line or even the one after. Using the reported value directly charges the torque of the previous line to later lines.

DTM's solution is **to decide by position**: a sample belongs to the line whose taught path the robot is actually on; the reported line number is used only to narrow the search range.

6.3 Hidden Markov model and the Viterbi algorithm

The expected path of each motion line (the curve in joint space from its start to its end) is represented by $K = 12$ points; two adjacent points form a small segment, and the small segments of all lines are arranged in program order into a state sequence $g = 1, 2, \dots, G$. Alignment then becomes: given an observation sequence $\mathbf{x}_1, \dots, \mathbf{x}_n$ (joint angles downsampled to 20 per second), find the most likely state sequence.

Observation model: let $d_g(\mathbf{x})$ be the distance from sample $\mathbf{x}$ to small segment g (projected onto the line segment, with the parameter u clipped to $[0, 1]$). Assuming the position deviation is isotropic and normal,

$$-\ln p(\mathbf{x} \mid g) = \frac{d_g(\mathbf{x})^2}{2\sigma^2} + \text{const}, \qquad \sigma = 1^\circ. \tag{6.1}$$

Transition model: the robot can only stay or move forward in program order, never backwards. Each small segment advanced costs c_s, crossing into the next line adds c_ℓ (equivalent to the negative

log of the transition probability), and at most $J = K - 1$ small segments can be advanced per step.

Viterbi recursion

Let $V_t(g)$ be the minimum cumulative cost of "being in state g at sample t":

$$V_t(g) = \frac{d_g(\mathbf{x}_t)^2}{2\sigma^2} + \min_{g-J \le h \le g} \left[V_{t-1}(h) + c_s\,(g-h) + c_\ell\,(\ell(g) - \ell(h)) \right] \tag{6.2}$$

$\ell(g)$ is the program line to which small segment g belongs. The h that gives the minimum is recorded at each step; tracing back from the end state with the lowest cost gives the best state sequence for the whole cycle. The computation is $O(n \cdot G \cdot J)$: dynamic programming replaces enumeration of all paths.

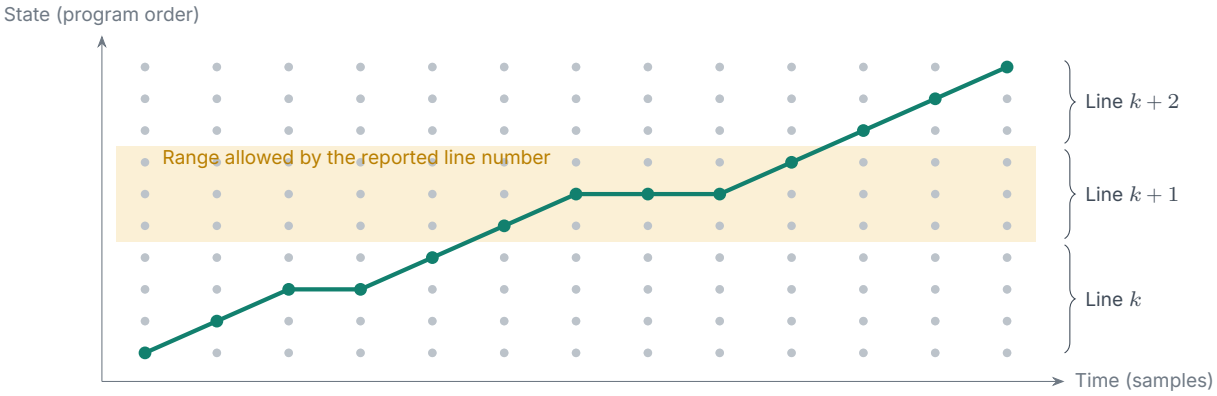


Figure 6.1 Viterbi alignment (schematic). Each row is a small path segment; the path can only go horizontally (stay) or upwards (advance). When line numbers are reported, the candidate states are limited to the range from "a few lines before the reported line" to "the reported line" (yellow band).

Details:

- **Learning path templates**: when line numbers are reported, an initial split is made with the reported values and the speed minima, a template is learned for each line, and the templates are refined repeatedly with the alignment results (similar to hard-assignment k-means iteration); without line numbers, the joint values of consecutive taught points are connected to form the initial templates.
- **Standstill periods**: when the norm of joint velocity is below $1\,°/\mathrm{s}$, the position does not change and the line cannot be decided by position. If a line change falls in a standstill period, it is moved to the moment the robot actually starts moving, and the standstill is counted as dwell of the previous line (waiting for a signal, positioning check, gripping).
- **Confidence**: the fraction of samples whose distance to the selected path after alignment is below 2σ ($2°$). Cycles with confidence below 0.6 are judged alignment failures and cycles passing through fewer than 90% of the states are judged incomplete; neither is included in the statistics.

6.4 Synchronous averaging: aligning cycles of different lengths

Waiting times differ from cycle to cycle, so aligning directly by time shifts the waveforms. After alignment, the samples within each line are resampled by in-line progress $u \in [0,1]$ (the projection parameter) onto 50 fixed grid points, and the median and the 10th and 90th percentiles are computed for each line and grid point, giving a "typical waveform" and a "normal range of variation". A rising coefficient of variation between cycles is itself a sign of an unstable line.

Abnormal cycles: a robust z-score is used so that abnormal cycles do not drag the reference themselves:

$$z_{\text{rob}} = \frac{|x - \operatorname{median} x|}{1.4826\,\mathrm{MAD} + 0.03\,|\operatorname{median} x|}, \qquad \mathrm{MAD} = \operatorname{median}_i\ |x_i - \operatorname{median}_j x_j| \tag{6.3}$$

For a normal distribution $1.4826\,\mathrm{MAD}$ is a consistent estimator of the standard deviation; 3% of the median is added to the denominator as a floor, for the same reason as in Section 3.2. When $z_{\text{rob}} > 4$ for motion time, damage or peak (after 20 cycles have accumulated), the cycle is marked "abnormal": it is still counted in the damage ledger and in improvement verification (load that really occurred should not be thrown away), but it does not take part in the typical waveform.

6.5 Per-line damage ledger

From Eq. (1.5), the damage and equivalent torque of each line ℓ are calculated from the discrete samples:

$$D_\ell = \sum_{i\in\ell} \frac{|\Delta\theta_i|}{360^\circ} \left(\frac{|T_i|}{T_0}\right)^p, \qquad T_{m,\ell} = \left(\frac{\sum_{i\in\ell} |\Delta\theta_i|\,|T_i|^p}{\sum_{i\in\ell} |\Delta\theta_i|}\right)^{1/p} \tag{6.4}$$

- Life fraction consumed per cycle $= \sum_\ell D_\ell / R_0$; daily consumption = consumption per cycle × cycles per day; estimated days until rated life is reached $= R_0 / (\sum_\ell D_\ell \times$ cycles per day) (see Section 1.3 for its meaning).
- The damage share of each line, $D_\ell / \sum D$, shows directly "which line is most worth changing". In the built-in demonstration data, the three motions on lines 12–14 account for about 80% of the damage in that period.
- When the angular velocity is close to zero (holding torque), Eq. (6.4) is zero, but small oscillations can still cause fretting wear; "holding time × torque" and the number of small swings are recorded separately (Section 6.6).

> **Use transmitted torque, not disturbance torque**
> T in Eq. (6.4) must be the total torque actually transmitted by the reducer (the torque command, or motor current × torque constant × reduction ratio × efficiency). Disturbance torque has already had gravity and inertia removed (Section 2.2); calculating with it yields only a relative ranking, and absolute life has no meaning. When only disturbance torque is available, DTM says so on screen and in the reports.

Peaks and stop events: for each line, the ratio of max $|T|$ to the "start/stop allowable torque" and the number of exceedances serve as the single-event check; events above the "momentary maximum allowable torque" are counted against the allowed number of occurrences C, each consuming $1/C$ of

the life. Stop events such as emergency stops, collisions and opening of the safety fence are counted from the controller alarm history and weighted by stop type (example: emergency stop $0.02/C$, collision detection $0.5/C$, controlled stop $0.005/C$).

6.6 Angle concentration map

The output angle of the reducer is divided into 1° bins k, and the damage density of each bin is accumulated:

$$d(\theta_k) = \sum_{i:\,\theta_i \in k} \frac{|\Delta\theta_i|}{360^\circ} \left(\frac{|T_i|}{T_0}\right)^p \tag{6.5}$$

The number of passes, the number of reversals and the amplitude of each swing are accumulated at the same time. The degree of concentration is quantified with three indicators:

- θ_{80}: the narrowest contiguous angle range containing 80% of the damage. If a joint actually uses 180° but $\theta_{80} = 20^\circ$, eighty percent of the wear is on the same small range of tooth flanks and raceways.
- **Gini coefficient**: with the damage of the bins sorted in ascending order $x_{(1)} \le \ldots \le x_{(n)}$,

$$\text{Gini} = \frac{2\sum_{i=1}^{n} i\, x_{(i)}}{n \sum_i x_i} - \frac{n+1}{n}, \tag{6.6}$$

  where 0 means perfectly even and values close to 1 mean concentration in a few bins.
- **Share of the top 5 bins**.

Small swings: reversal detection uses a hysteresis (0.05°) to exclude noise; the number of swings smaller than the contact width (example 0.5°) is recorded as fretting-wear risk (Section 1.6). When the reduction ratio and the numbers of cycloidal-disc teeth and pins are entered, the output angle can be converted to the corresponding pin or raceway position and compared with the damage location seen at teardown.

6.7 Full-range sweep and the local wear index

The angle concentration map only shows "where the load was highest", not "that this place is already worn". The full-range sweep provides the evidence: once a week, each axis runs slowly forward and in reverse over the largest interference-free range, and the friction $F(\theta_k)$ of each angle bin is calculated with Eq. (2.6). The angle concentration map is used to define:

- frequently used region U: bins accumulated in order of decreasing damage density until 50% is reached;
- reference region R: bins that were passed but have the lowest 30% damage density.

$$\text{LWI} = \frac{\text{median}_{k\in U}\, F(\theta_k)}{\text{median}_{k\in R}\, F(\theta_k)} \tag{6.7}$$

Why this is strong evidence: U and R belong to **the same reducer and the same sweep**, with exactly the same temperature, grease and payload setting, so all common factors cancel in the ratio.

If wear is local, friction in the frequently used region rises while the reference region does not, and LWI rises; if only the grease or temperature changes, both regions change together and LWI does not. This is a measurement design "with a built-in control group". LWI ≥ 1.2 times the first sweep means caution and ≥ 1.4 times alert.

Running slowly over a wide range also redistributes grease into rarely used sections, which may reduce the wear caused by small swings (effect still to be verified).

DTM implementation

- The instrumented program only inserts `R[114]=`line number before each motion line; motions, speeds and points are unchanged. When determining the "effective positioning type", a CNT immediately followed by WAIT, CALL or END is treated as FINE (the controller actually stops there).
- Alignment: 12-point path templates, downsampling to 20 Hz, $\sigma = 1^\circ$; with line numbers, at most 2 lines back are searched, crossing only CNT lines; with positions only, chord-length templates and beam search are used.
- Angle bins 1° (10 mm for linear axes); reversal hysteresis 0.05° (0.5 mm); small-swing threshold 0.5°.
- The sweep program marks its segments with the segment code $8000+$ axis $\times 10+$ direction, which are separated automatically during streaming or import.

7 Load reduction without affecting cycle time: time-scaling theory

Once the program lines that wear the reducer have been found, the most direct remedy is to lower the speed or acceleration of those lines, but that usually lengthens the cycle time, which production will not accept. This chapter explains the core idea of DTM — **use the time the robot already spends waiting** to slow motions down, so that the load falls while the cycle time stays the same — and derives how the effect can be estimated from measured data.

7.1 Wait slack

After many motions the robot stops and waits for an external signal (fixture in place, conveyor, next station finished). Let the time of a motion block (the motions from the previous wait up to this wait) be T_{move} and the actual waiting time per cycle be w. As long as the slowed robot still arrives before the signal comes, the cycle time does not change. The usable slack is taken conservatively:

$$S = Q_q(w) - T_{\text{timer}} - \delta, \tag{7.1}$$

where $Q_q(w)$ is the q-th percentile of the actual waiting time (example $q = 10$: nine out of ten cycles wait longer than this), T_{timer} is fixed time such as timed waits, and δ is a safety margin.

7.2 The time-scaling theorem

Stretch the motion uniformly in time by a factor σ:

$$\mathbf{q}_\sigma(t) = \mathbf{q}(t/\sigma), \qquad \sigma = \frac{T_{\text{move}} + S}{T_{\text{move}}} \ (\geq 1). \tag{7.2}$$

The path is exactly the same; only the speed at which it is travelled changes. Differentiating with respect to time:

$$\dot{\mathbf{q}}_\sigma(t) = \frac{1}{\sigma}\,\dot{\mathbf{q}}(t/\sigma), \qquad \ddot{\mathbf{q}}_\sigma(t) = \frac{1}{\sigma^2}\,\ddot{\mathbf{q}}(t/\sigma). \tag{7.3}$$

Derivation: torque after scaling (Hollerbach, 1984)

Substituting into Eq. (2.1) and comparing at corresponding times (the same path position):

$$\tau_\sigma = \frac{1}{\sigma^2}\underbrace{\left[\mathbf{M}(\mathbf{q})\ddot{\mathbf{q}} + \mathbf{C}(\mathbf{q},\dot{\mathbf{q}})\dot{\mathbf{q}}\right]}_{\text{inertia and Coriolis: } \propto 1/\sigma^2} + \underbrace{\mathbf{g}(\mathbf{q})}_{\text{gravity: unchanged}} + \underbrace{\tau_f(\dot{\mathbf{q}}/\sigma)}_{\text{friction: with speed}} \tag{7.4}$$

Coriolis and centrifugal forces are quadratic in velocity, so like the inertia term they are divided by σ^2; gravity depends only on position and is unaffected; viscous friction is divided by σ and Coulomb friction is unchanged.

7.3 Invariance of damage to the time parameterization

The angle form of Eq. (1.5) leads to an important conclusion:

$$D = \frac{1}{2\pi}\int_{\text{path}} \left(\frac{|T|}{T_0}\right)^p |\mathrm{d}\theta| \tag{7.5}$$

The integral runs along the **path** and does not depend on how long it takes to travel that path. Therefore:

Conclusion

Slowing a motion does **not** reduce the number of revolutions, and life is not extended because "fewer revolutions are made per second"; the reduction in damage comes **entirely** from the reduction in torque. The inertia term falls with $1/\sigma^2$ and is then amplified by the $10/3$ power. The higher the inertial share of a motion, the larger the effect of slowing it; motions dominated by gravity (for example moving slowly while holding a heavy load) gain almost nothing.

If the torque magnitude of a motion segment can be simplified as "a gravity part $(1-\phi)$ plus an inertia part ϕ" (original torque taken as 1, both parts of the same sign), the damage ratio after scaling is

$$\frac{D_\sigma}{D} = \left[(1-\phi) + \frac{\phi}{\sigma^2}\right]^{10/3}. \tag{7.6}$$

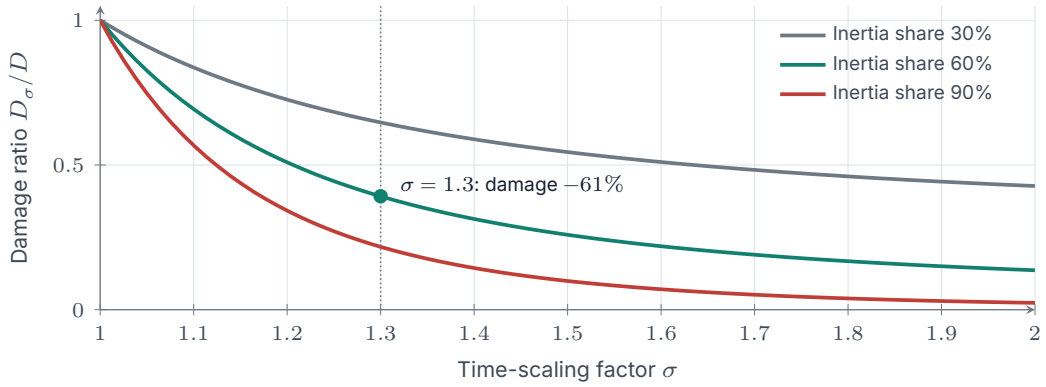


Figure 7.1 Damage ratio under time scaling (Eq. (7.6), schematic). The higher the inertial share, the larger the effect of slowing down.

Numerical example

Motion time 1.0 second, measured wait slack 0.3 second, $\sigma = 1.3$: acceleration and inertial torque become $1/1.3^2 \approx 0.59$ times and viscous friction $1/1.3 \approx 0.77$ times. If 60% of the torque in this segment comes from inertia, damage becomes about 0.39 times the original (-61%); if it is all inertia, $1.3^{-20/3} \approx 0.17$ times. The robot still arrives before the signal comes, so the cycle time does not change.

7.4 Estimating the scaled torque from measured data

Eq. (7.4) requires knowing how much of the torque comes from inertia, gravity and friction. DTM does not simulate with a dynamic model of the robot (the model parameters are unknown, and a simulation cannot reflect the actual payload, temperature and grease condition). Instead, it regresses each line and each axis over many **measured** cycles:

$$\tau(u) = \underbrace{a_0 + a_1 u + a_2 u^2 + a_3 u^3}_{\text{depends only on in-line position (gravity etc.)}} + c\,\alpha + v\,\omega + e \tag{7.7}$$

where u is the in-line progress, α the angular acceleration and ω the angular velocity. Because the path is fixed, position is a function of u alone, and the cubic polynomial absorbs all terms that depend only on position (gravity, balancer); c estimates the equivalent inertia and v the viscous coefficient. The acceleration and velocity differ slightly from cycle to cycle, which provides the variation needed to estimate c and v. The estimated torque after scaling is

$$T' = T - c\,\alpha\left(1 - \frac{1}{\sigma^2}\right) - v\,\omega\left(1 - \frac{1}{\sigma}\right), \tag{7.8}$$

and recalculating the damage with Eq. (6.4) gives "how much damage per cycle falls after slowing down".

Assumptions and limitations

- **The path does not change**: the controller's CNT blending depends on speed, so changing the speed may change the corner path slightly. The suggestion is text only; a person must check interference with simulation software and run a low-speed trial.
- **Uniform scaling**: actual changes usually lower the speed or acceleration override, which is not exactly uniform time scaling; after the change, the result should be verified by measurement with a post-change diagnosis (Section 8).
- Estimation uses measured data only; DTM does not estimate load by simulating the robot's motion from the program, nor does it change programs or reduce speed automatically.

7.5 Shape of the acceleration profile

For the same distance D_s and time T, rest to rest, different velocity profiles have different peak accelerations. In dimensionless time $\tilde{t} = t/T$ and dimensionless displacement $s(\tilde{t}) = x/D_s$:

Derivation: peaks of three profiles

Square-wave acceleration (bang–bang): accelerate for the first half, decelerate for the second; $v_{\max} = 2D_s/T$, $a = 4\,D_s/T^2$. This is the profile with the smallest peak acceleration, but the acceleration jumps instantaneously at the start, the end and the midpoint (infinite jerk).
Trapezoidal velocity (acceleration, constant speed and deceleration 1/3 each): $v_{\max} \cdot \frac{2T}{3} = D_s \Rightarrow v_{\max} = 1.5\,D_s/T$; $a = v_{\max}/(T/3) = 4.5\,D_s/T^2$.
Minimum jerk (Flash & Hogan, 1985): the fifth-order polynomial that minimizes $\int \dddot{x}^2\,\mathrm{d}t$,

$$s(\tilde{t}) = 10\tilde{t}^3 - 15\tilde{t}^4 + 6\tilde{t}^5, \qquad s''(\tilde{t}) = 60\tilde{t} - 180\tilde{t}^2 + 120\tilde{t}^3.$$

Setting $s''' = 0$ gives $\tilde{t} = (3 - \sqrt{3})/6$, and substituting gives the peak $s''_{\max} = 10/\sqrt{3} \approx 5.77$, that is $a_{\max} = 5.77\,D_s/T^2$; the peak velocity is $1.875\,D_s/T$.

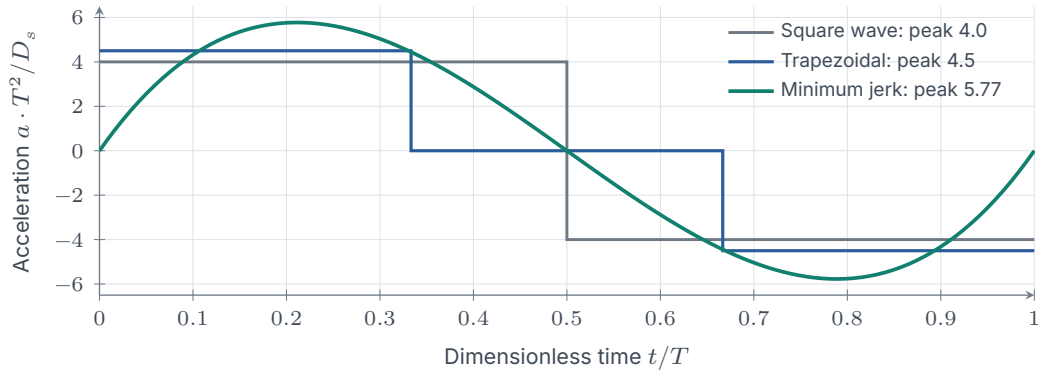


Figure 7.2 Acceleration of three rest-to-rest profiles. The smoother the profile, the higher its peak in the same time.

Table 7.1 Comparison of the three profiles (same distance, same time)

	Square wave	Trapezoidal (1/3)	Minimum jerk
Peak acceleration $\kappa = a_{\max} T^2/D_s$	4.00	4.50	5.77
Peak velocity $v_{\max} T/D_s$	2.00	1.50	1.875
Jerk	Infinite	Infinite (at corners)	Finite, smooth
Cumulative damage, inertia only (relative)	101.6	75.2	128.8

The last row of Table 7.1 is the numerical integral $\int |s''|^{10/3} |\mathrm{d}s|$ from Eq. (1.5) under the assumption of **inertial torque only** ($T \propto a$). It shows a counterintuitive fact: **a smoother profile is not necessarily gentler on the reducer**. The minimum-jerk profile has the smallest impacts, but its peak acceleration is higher and its damage accumulated with the $10/3$ power is actually the largest; the trapezoidal profile has no inertial torque during its constant-speed phase and the smallest cumulative damage.

DTM's decision logic is therefore:

- **Impacts and resonance** (visible with an accelerometer, high crest factor): reduce jerk, for example by adding intermediate points for a soft start or lowering the acceleration override.
- **Cumulative damage** (high in the $10/3$-power ledger): reduce peak acceleration.
- **Wait slack available**: simply lengthen the time; peak and jerk fall together and both problems are solved at once.

In real data, DTM uses $\kappa = a_{\max} T^2/D_s$ to judge which profile each rest-to-rest motion is closest to.

7.6 Without slack: trading cycle time for life

For lines on the critical path with no wait slack, slowing down increases the cycle time. For these lines DTM lists a trade-off table: for each additional 0.1 second ($\sigma = (T + 0.1)/T$), the reduction in damage and the extension of expected life estimated with Eq. (7.8), converted into changes in annual replacements and downtime hours. Whether it is worth it is decided by production and

maintenance on the basis of cost:

$$\text{annual saving} = N_{\text{avoided}} \times \left(H_{\text{downtime}} \times C_{\text{per hour}} + C_{\text{spare}} + C_{\text{labor}}\right) + C_{\text{output loss before failure}} \tag{7.9}$$

DTM implementation

- Wait slack S = 10th percentile of the actual waiting time − timed waits − 0.05 second − safety margin (example 0.3 second); the upper limit of σ is 2.
- Each line and axis is regressed with Eq. (7.7); for lines without slack, the trade-off is calculated per additional 0.1 second.
- Change suggestions (rule-based): change FINE to CNT at points that do not need precise positioning; lower ACC or speed on lines with high damage and slack; add intermediate points for a soft start on long motions; avoid unnecessary large wrist flips; check the payload setting; avoid linear motions through high-torque postures where a joint motion will do. Suggestions are output as text only, and a person makes the changes.

8 Verification: proving that improvements work and that decisions are accurate

Reducer life is measured in years; waiting for the next failure to see whether an improvement worked takes far too long, and before/after differences may also come from changes in product, season or output. This chapter explains two kinds of verification: improvement verification (did the program change reduce the load?) and field validation (are the monitoring decisions accurate?).

8.1 Difference in differences

One period is measured before the change and one after; in addition, a robot of **the same model, with similar load and not modified** is chosen as the control and measured during both periods. The control experiences the same changes in environment, product and season, so its "after ÷ before" represents the change unrelated to the modification.

$$\text{effect} = \frac{\bar{x}^{\mathrm{t}}_{\text{after}}/\bar{x}^{\mathrm{t}}_{\text{before}}}{\bar{x}^{\mathrm{c}}_{\text{after}}/\bar{x}^{\mathrm{c}}_{\text{before}}} - 1 = \exp\left[(\overline{\ln x}^{\mathrm{t}}_{\text{after}} - \overline{\ln x}^{\mathrm{t}}_{\text{before}}) - (\overline{\ln x}^{\mathrm{c}}_{\text{after}} - \overline{\ln x}^{\mathrm{c}}_{\text{before}})\right] - 1 \qquad (8.1)$$

The superscript t denotes the treated robot (the one modified) and c the control; x is the median of each period (daily, hourly or every 10 minutes), and averages are geometric means (means of logarithms).

Why logarithms: changes in quantities such as damage and torque are **multiplicative** ("30% less") and their distributions are right-skewed. Logarithms turn ratios into differences, so that the difference in differences becomes additive, and they also give periods with large and small values similar weight.

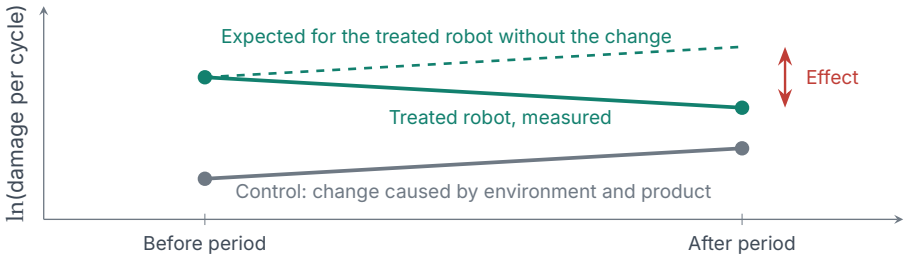


Figure 8.1 Difference in differences (schematic). The actual change of the treated robot minus the change of the control over the same period is the effect of the modification.

Assumption: parallel trends, that is, without the modification the treated and control robots would have changed in the same way. The control should therefore be of the same model, with similar load, measured over the same periods. Without a control, DTM still calculates after ÷ before but states in the report that "changes in environment and product cannot be removed".

8.2 Bootstrap confidence intervals

The effect in Eq. (8.1) is a non-linear statistic built from several medians, with no simple formula for its standard error. The bootstrap (Efron 1979) estimates its sampling distribution directly by

resampling:

1. **Within each** of the four groups (treated before, treated after, control before, control after), resample the same number of periods with replacement.
2. Calculate the effect once for the resampled data.
3. Repeat 2000 times and take the 2.5th and 97.5th percentiles as the 95% confidence interval.

Periods rather than cycles are resampled because cycles within the same day are correlated (same batch of products, same temperature); treating them as independent samples would make the interval too narrow. If the interval does not contain 0, the effect is statistically credible.

8.3 Mann–Whitney test

In addition, a non-parametric test is applied to the treated robot before and after, with no assumption of normality. The two groups are pooled and ranked, and the rank sum R_1 of the first group is calculated:

$$U = R_1 - \frac{n_1(n_1+1)}{2} \tag{8.2}$$

For small samples ($n_1 n_2 \le 400$) without ties, the exact distribution of U is computed by dynamic programming; otherwise the normal approximation is used with a tie correction:

$$\mu_U = \frac{n_1 n_2}{2}, \qquad \sigma_U^2 = \frac{n_1 n_2}{12}\left[(N+1) - \frac{\sum_k (t_k^3 - t_k)}{N(N-1)}\right], \qquad z = \frac{|U-\mu_U| - 0.5}{\sigma_U} \tag{8.3}$$

where $N = n_1 + n_2$, t_k is the size of the k-th group of ties, and 0.5 is the continuity correction.

> **Results with the demonstration data**
>
> In the built-in demonstration, R01 changes line 13 from FINE to CNT80, uses wait slack to slow lines 6, 24 and 25, and adjusts P[7] to avoid a wrist flip; R02 is the control. Result: cycle time $+0.1\%$ (unchanged), J1 damage per cycle -36%, while J2 damage rises slightly ($+6.7\%$), which is reported as well. Improvement verification does not report only the good results; all indicators have to be looked at.

8.4 Field validation: detection rate and Wilson intervals

Whether the monitoring decisions are accurate must ultimately be checked against teardown results. For each confirmed failure (damage found at teardown on replacement), DTM looks back over a period before the replacement (example 365 days) to see whether a caution or alert had been issued:

- **Detection rate** = failures with a prior warning / total failures;
- **Days of warning**: days from the first warning to the replacement;
- **Missed**: failures without a warning; **insufficient data**: fewer than 30 days of valid data in the 90 days before the failure, not counted in the denominator;
- **Possible false alarms**: alerts with no maintenance or inspection at all within 90 days, or alerts resolved only by a "baseline reset" (the cause was not component deterioration).

The detection rate is a proportion and the samples are small, so a confidence interval must be given. The common $\hat{p} \pm 1.96\sqrt{\hat{p}(1-\hat{p})/n}$ is very inaccurate when n is small or $\hat{p}$ is close to 0 or 1 (it can even go outside $[0, 1]$); DTM uses the Wilson interval:

$$\frac{\hat{p} + \dfrac{z^2}{2n} \pm z\sqrt{\dfrac{\hat{p}(1-\hat{p})}{n} + \dfrac{z^2}{4n^2}}}{1 + \dfrac{z^2}{n}}, \qquad z = 1.96 \tag{8.4}$$

Table 8.1 95% Wilson intervals for the same detection rate with different sample sizes

	9/10	18/20	45/50
Detection rate	90%	90%	90%
95% confidence interval	60%–98%	70%–97%	79%–96%

Detecting 9 of 10 failures may mean a true detection rate of only 60%. This is why this paper stresses repeatedly that thresholds can only be confirmed after enough teardown records have accumulated, and that teardown results of "no abnormality found" are equally important.

8.5 Threshold replay

Before thresholds are changed, another set of thresholds and evidence weights can be applied to the stored diagnostic features and daily medians, the level of each run (each day) recalculated, and the same hysteresis rule applied. The output, "how many cautions and alerts the old and new settings would each issue, and when the first one comes", together with the teardown results of Section 8.4, lets the pros and cons of a threshold change be evaluated without waiting for new failures: lower thresholds find problems earlier but also produce more false alarms.

DTM implementation

- The statistical unit is chosen according to the amount of data: daily, hourly or 10-minute medians (a longer unit is used only if each group has at least 3 periods).
- Bootstrap with 2000 repetitions; Mann–Whitney uses the exact distribution when $n_1 n_2 \le 400$ and there are no ties.
- Field validation report: looks back 365 days (but not before the component installation date); a failure can be evaluated only if there are at least 30 days of monitoring data in the 90 days before it; alerts on the same axis separated by no more than 3 days count as one; maintenance already scheduled within the 7 days before an alert also counts as action taken; replacements found "no abnormality" at teardown are not counted as failures but listed as "replaced after alert, failure not confirmed".

9 Assumptions, limitations and validation status

How far a method can be trusted depends on whether its users know clearly under what conditions it holds. This chapter collects the assumptions and known limitations of the previous chapters and describes the current validation status of v2.1.0.

9.1 Model assumptions

Table 9.1 Main assumptions and the consequences when they fail

Assumption	Used in	When it does not hold
Disturbance torque ≈ friction + model error (Eq. (2.2))	Friction decomposition	With a different controller algorithm the meaning of F and G may change; at commissioning, first confirm on the teach pendant that the variables match the screen values
Friction is nearly linear over the diagnostic speed range (Eq. (2.4))	F_c, F_v regression	At low speed in the Stribeck region F_c is overestimated; more speed points or a non-linear model can be used
Friction varies exponentially with temperature, with a fixed coefficient	Temperature compensation	Coefficients differ between greases; estimating them from your own data with Eq. (3.6) is recommended
Fewer than half of the axes are deteriorating	Common change across axes	Simultaneous deterioration of all axes is removed; grease life and iron debris guard against this
Evidence families are conditionally independent	Two-evidence rule	Grease deterioration affects both torque and iron debris, which are therefore not fully independent; family coefficients should be corrected with teardown results
Miner's linear accumulation and the catalog power law	Damage ledger, life estimation	Load-sequence effects and lubrication-failure damage do not fully fit the fatigue model; absolute life needs calibration with replacement records, and relative rankings are more reliable
Components of the same type share a life distribution	Fleet Weibull prior	Plants with very different operating conditions should be fitted separately
The path does not change under time scaling	Wait-slack estimate	CNT corners change with speed; changes must be checked for interference by a person and verified with a post-change diagnosis
Parallel trends	Difference in differences	If the control and treated robots differ in conditions, the effect estimate is biased

9.2 Where the thresholds and coefficients come from

The thresholds in this paper (for example ratios 1.5/2.0, production monitoring 1.15/1.30, iron debris 0.05/0.10 wt%), the evidence coefficients (0.45, 0.8, 0.6) and the temperature coefficients (0.025, 0.004/°C) are all **starting values**, based on the author's roughly ten years of FANUC robot main-

tenance experience and public selection data, not derived statistically from large amounts of field data. Their purpose is to let the system work without historical data; the real thresholds are to be corrected step by step with the field validation report of Section 8.4 and the threshold replay of Section 8.5.

9.3 Validation status of v2.1.0

Beta: not yet connected to a real controller and not yet validated on a production line

- All computer-side methods are implemented and have been run end to end with 233 automated tests and the built-in simulator and demonstration data.
- The robot-side programs (background logic, diagnostic program, full-range sweep, KAREL streaming) are templates; variables, read rates and motions have not yet been confirmed on a real robot.
- The decision methods in this paper have not yet been compared with teardown records from a production line; detection and false-alarm rates are unknown.

Start with a 1–3 month trial in "shadow mode": observe only, do not base replacements on it, compare it in parallel with the existing maintenance practice, and fill in the teardown form for every replacement. v3.0 will be released after the first validation on a real robot. The completion and verification level of each feature is in the *Feature Status* table in the source code.

Signal checks before commissioning

The first premise of the whole method is that Eq. (2.2) holds for the signals actually read. The controller's internal estimation method, units and filtering are not published, so at commissioning the check has two parts:

Manual checks (once per controller and axis)

1. The value read matches the value shown on the teach pendant (same moment, same axis).
2. Unit, scale and sign have been confirmed; if uncertain, use the values only as relative values, do not convert them to N m and do not estimate life.
3. The controller software version and options have been recorded; controllers with different versions or options are checked separately.
4. The physical response is plausible: run the diagnostic program once cold and once after warm-up; F_v after warm-up should be clearly lower. If the direction is reversed or there is no change, the signal being read is not dominated by friction, and the cause must be found before using it.

Automatic checks by the program (commissioning report: produced automatically after 7 days of running, and available at any time with the `commissioning-report` command): actual sampling rate and jitter (P95, P99), data gaps and connection, success rate of diagnostic segment cutting and reasons for discards, noise of each axis (ripple, coefficient of variation of F), coverage of motion codes. Items marked as warnings in the report must be dealt with before maintenance is scheduled on the basis of DTM's decisions.

9.4 Known limitations and planned corrections

The following issues were pointed out by external review before the release of v2.1.0, confirmed by the author, and will be addressed in the next version. They do not affect the correctness of the formulas and numerical examples in this paper, but they do affect how some results should be read:

- **Damage ledger with disturbance torque only** (Section 6.5): disturbance torque does not contain gravity and inertia, so the accumulated result reflects friction work (a tendency to wear), not a ranking of fatigue damage. The next version will call it a "friction-work hotspot" and use $p = 1$.
- **Angle concentration map** (Section 6.6): it shows load hotspots in terms of the reducer's **output angle**. While the output of an RV reducer turns 20°, the crankshafts have already turned several hundred degrees, so the pins and crankshaft bearings are loaded almost all the way round; what can be mapped directly to a local position is mainly the raceway of the rotating ring of the main bearing, and fretting wear from small swings. "Tooth flanks and raceways" in the text should be understood in this sense.
- **Iron-debris turning point** (Section 4.5): reaching the caution concentration is a screening signal of wear risk, not confirmation that the accelerating stage has begun. A grease change is not a complete replacement; if no sample is taken after the change, the generation rate of the next batch will come out too high.
- **Wait slack** (Section 7): the 10th percentile plus a safety margin only lowers the risk of affecting cycle time and is not a guarantee. The per-line regression does not include multi-axis coupling terms and has no quality threshold or error range yet, so its results should be treated as exploratory estimates and post-change measurements take precedence; the most direct check is a trial run at $1/\sigma$ of the speed with measurement.
- **Improvement verification** (Section 8.2): independent resampling within each group does not account for day-to-day autocorrelation of the same robot, so the interval may be too narrow; a date-paired block bootstrap is planned. The Mann–Whitney test is used only as a supplement, not as causal evidence.
- **Life statistics** (Section 5): treating parts replaced early after an alert and found undamaged at teardown as censored data biases η optimistically. Uncertainty intervals for β and η (obtainable directly from the profile likelihood) are planned.
- **Torque conversion**: a reducer's efficiency differs when it drives and when it is back-driven, so converting to transmitted torque with a single coefficient gives errors in gravity-driven segments, amplified by the $10/3$ power; friction of the J2 balancer of the R-2000iC also gets into F. Both need data from a real robot to resolve.
- **Life exponent**: $p = 3$ for wrist axes and gearboxes is an assumed value; the catalog name and edition will be added to reference [18].

9.5 What is deliberately not done

For reasons of field safety, credibility of decisions and patent screening, this project chooses not to implement the following:

- The computer side (DTM) does not write data to the controller or send motion commands; the robot-side templates (background logic, diagnostic program, motion codes, instrumented program) are installed by people and write only to dedicated registers (for example R[110], R[114]). Change suggestions are output as text only; programs are not modified and speeds are not reduced automatically.
- Loads simulated from the program are not used in place of measured data for decisions; the built-in simulator only produces demonstration data for the built-in demonstration program.
- Replacement of a reducer or motor is not recommended on a single kind of evidence. The only exception is age risk: when the conditional failure probability from the Weibull life model is high enough, preventive replacement is recommended on the basis of life statistics even without an abnormal signal (Section 4.8).
- No millisecond-level spectrum analysis; where needed, use the manufacturer's tools or dedicated equipment.

The detailed reasons are in the *Design Constraints* in the source code; the technical comparison with known patents is in Section 6.15 of the *Technical Disclosure*.

9.6 Conclusion

The DTM method can be condensed into three sentences:

(1) **Measure the right thing**: take friction from disturbance torque by subtracting forward and reverse runs, and calculate damage with transmitted torque; do not mix the two.

(2) **Compare with itself, remove common factors, require two kinds of evidence**: reduce false alarms and avoid mistaking temperature, season or setting errors for wear.

(3) **Damage accumulates along the path with the $10/3$ power**: find the program lines and angles that wear the reducer, use waiting time to slow motions down, and prove the effect with measurements against a control.

Every step has explicit assumptions, and every step can be tested and corrected with field data. This document is published together with the code under Apache-2.0; checks against real data and corrections are welcome.

Appendix A Example parameters

Table A.1 Example parameters mentioned in this paper (adjustable in the configuration file)

Category	Parameter	Example value
Sampling	Continuous monitoring; Pro diagnostic period; accelerometer	2–5 per second; 50–125 per second; 5–25 kHz
Diagnostic program	Speeds; reference speed; end trimming; minimum constant-speed time	5/10/20%; 10%; 1.0 s; 2.0 s
Baseline	Number of runs; runs required for z; standard-deviation floor	10; 10; 5%
Decision	Ratio caution/alert; z caution/alert; minimum ratio for early escalation; consecutive confirmations	1.5/2.0; 2/3; 1.15; 2
G offset	$\lvert G - G_{\text{base}}\rvert / F_{\text{base}}$ caution/alert	0.5/1.0
Temperature compensation	k_v; k_c; temperature spread needed to learn the coefficient	0.025/°C; 0.004/°C; 5 °C
Common change	Limit of equivalent temperature difference; time difference within one diagnostic run	±30 °C; 1 hour
Production monitoring	Valid cycles per day; baseline days; caution/alert; confirmation	≥ 20; 10 days; 1.15/1.30; 2 days
Trend	Window; acceleration ratio; quadratic residual ratio; extrapolation limit; range	30 runs; 1.5; 0.8; 730 days; ±(30% + 7 days)
Vibration	Ratio caution/alert; kurtosis; axial ratio	2/3; 4/6; 1.5
Iron debris	Concentration caution/alert (RV); W ratio; minimum operating hours; sampling interval	0.05/0.10 wt%; 2/3; 300 h; 60/21 days
Temperature	Reference temperature; sudden-rise threshold; exclusion after grease change	40 °C; +8 °C or +5 °C/day; 72 h
Decisions	Single-family threshold and coefficient; action/now	0.25, 0.45; 0.5/0.9
Life	Minimum failures for Weibull; spare lead time; conditional failure probability	3; 60 days; 0.2
Damage	Life exponent p; start/stop and momentary maximum allowable	10/3 (RV), 3 (gearbox); 2.5/5 times rated
Stop events	Emergency stop / collision / controlled stop (each)	$0.02/C$, $0.5/C$, $0.005/C$
Alignment	Template points; alignment rate; position noise σ; maximum lines back	12; 20 Hz; 1°; 2 lines
Angle concentration	Bin width; reversal hysteresis; small-swing threshold	1°; 0.05°; 0.5°
Sweep	LWI caution/alert; frequently used / reference region	1.2/1.4; 50%/30%
Wait slack	Percentile; fixed deduction; safety margin; upper limit of σ	10th; 0.05 s; 0.3 s; 2
Improvement verification	Bootstrap repetitions; condition for the exact test	2000; $n_1 n_2 \leq 400$
Field validation	Look-back; coverage window / minimum days; merge gap	365 days; 90/30 days; 3 days

Appendix B References

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[19] FANUC CORPORATION. *ZDT* (*Zero Down Time*) product catalog ZDT(E)-05. `https://www.fanuc.co.jp/en/product/catalog/pdf/robot/ZDT(E)-05.pdf`

About this paper

Author: Domingo He. This paper explains the decision methods of DTM v2.1.0 (Beta). The concept, decision rules and field knowledge were provided by the author; the text, the organization of the formulas and the program code were written with the help of AI (Anthropic Claude) and confirmed by the author (I design, AI writes the code, I verify). All numerical examples in this paper are illustrative or come from the software's built-in demonstration data and contain no actual production data of any company. Monitoring results are only an input to maintenance decisions; decisions on replacement, stopping and safety must be made by qualified personnel according to the manufacturer's manuals and the situation on site. FANUC is a trademark of FANUC CORPORATION; this project has no affiliation with FANUC.

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